Third Party Cloud Services: MetaCDN, SpotCloud, Federated Clouds / Inter Cloud

5.1

Third Party Cloud Services

- One of the key elements of cloud computing is **the possibility of composing services that belong to different vendors or integrating them into existing software systems.**
- The service-oriented model, which is the basis of cloud computing, facilitates such an approach and provides the opportunity for **developing a new class of services that can be called third-party cloud services.**

Third Party Cloud Services

- These are **the result of adding value to preexisting cloud computing services, thus providing customers with a different and more sophisticated service.**
- **Added value can be either created by smartly coordinating existing services or implementing additional features** on top of an existing basic service.

Third Party Cloud Services

Some examples of third-party services:-

- MetaCDN
- SpotCloud

What is a CDN?

- A content delivery network (CDN) is **a network of interconnected servers that speeds up webpage loading** for data-heavy applications.
- CDN can stand for content delivery network or content distribution network.

https://aws.amazon.com/what-is/cdn/

What is a CDN?

- When a user visits a website, data from that website's server has to travel across the internet to reach the user's computer. **If the user is located far from that server, it will take a long time to load a large file, such as a video or website image.**
- Instead, the **website content is stored on CDN servers geographically closer to the users and reaches their computers much faster.**

https://aws.amazon.com/what-is/cdn/

Why is a CDN important?

- The primary purpose of a content delivery network (CDN) is to **reduce latency, or reduce the delay in communication created by a network's design**.
- Because of the global and complex nature of the internet, **communication traffic between websites (servers) and their users (clients) has to move over large physical distances**.
- The communication is also two-way, with requests going from the client to the server and responses coming back.

https://aws.amazon.com/what-is/cdn/

Why is a CDN important?

- A CDN improves efficiency by introducing **intermediary servers between the client and the website server.**
- These CDN servers manage some of the client-server communications. **They decrease web traffic to the web server, reduce bandwidth consumption,** and improve the user experience of your applications.

https://aws.amazon.com/what-is/cdn/

CDN?

- You could think of a CDN like an ATM.
- Having a cash machine on practically every corner makes it fast and efficient to get money. **There's no wait time in long bank lines, and the ATMs are placed in many convenient locations for immediate access.**

https://www.akamai.com/our-thinking/cdn/what-is-a-cdn

CDN?

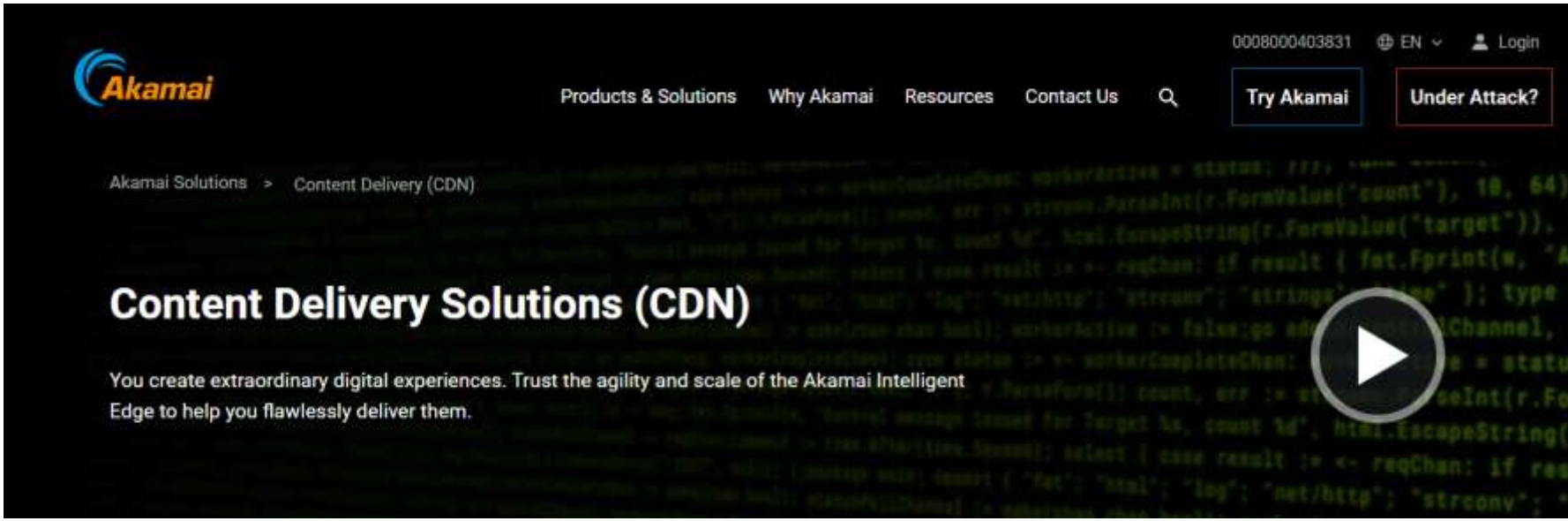
- CDN services were created to **solve the problem of network congestion caused by delivering rich web content, such as graphics and video over the internet — much like a traffic jam.**
- Getting content from centrally located servers to individual users simply took too long.
- **CDNs have now grown to include everything from text, graphics, scripts, and media files to software downloads, documents, portals, ecommerce, live streaming media, on-demand video streaming media, and social media sites.**

https://www.akamai.com/our-thinking/cdn/what-is-a-cdn

Existing CDN providers

- Akamai is the clear leader in coverage and market share (approx. 80%)
  - Price is prohibitive for SME, NGO, Gov...

- Academic CDNs include Coral, Codeen, Globule....



MetaCDN

- MetaCDN provides users with a Content Delivery Network (CDN).
- It implements a software overlay that **coordinates the service offerings of different cloud storage vendors and uses them as distributed elastic storage on which the user content is stored.**

- MetaCDN **enables the uniform use of heterogeneous storage clouds as a single, large, distributed content delivery network.**
- **The advantage is not only given by providing a CDN service at accessible costs but also in enriching the original service offering of existing cloud services with additional functionalities,**

- MetaCDN provides users with the **high-level services of a CDN** for content distribution and **interacts with the low-level interfaces of storage clouds to optimally place the user content in accordance with the expected geography of its demand.**

- MetaCDN is a pioneering "4th generation" content delivery network (CDN) provider. Our patent pending approach leverages **Tier-1 Cloud Storage & CDN suppliers such as Amazon, Microsoft and Google to offer enterprise-class content delivery, video encoding and streaming services on an unmatched global scale.**

http://www.metacdn.com/aboutus

- Founded in 2011 from **research out of the University of Melbourne**,
- MetaCDN is **backed by leading Australian venture capital firm Starfish Ventures and the University of Melbourne's commercialisation arm**

http://www.metacdn.com/aboutus

What if we could create a low-cost, high performance overlay CDN using heterogenous storage clouds?

- **Exploit the strengths and coverage footprints of multiple providers**
- **Cherry pick providers based on customers QoS needs / cost budget**
- **MetaCDN provides this while hiding the complexity from users**

MetaCDN Architecture

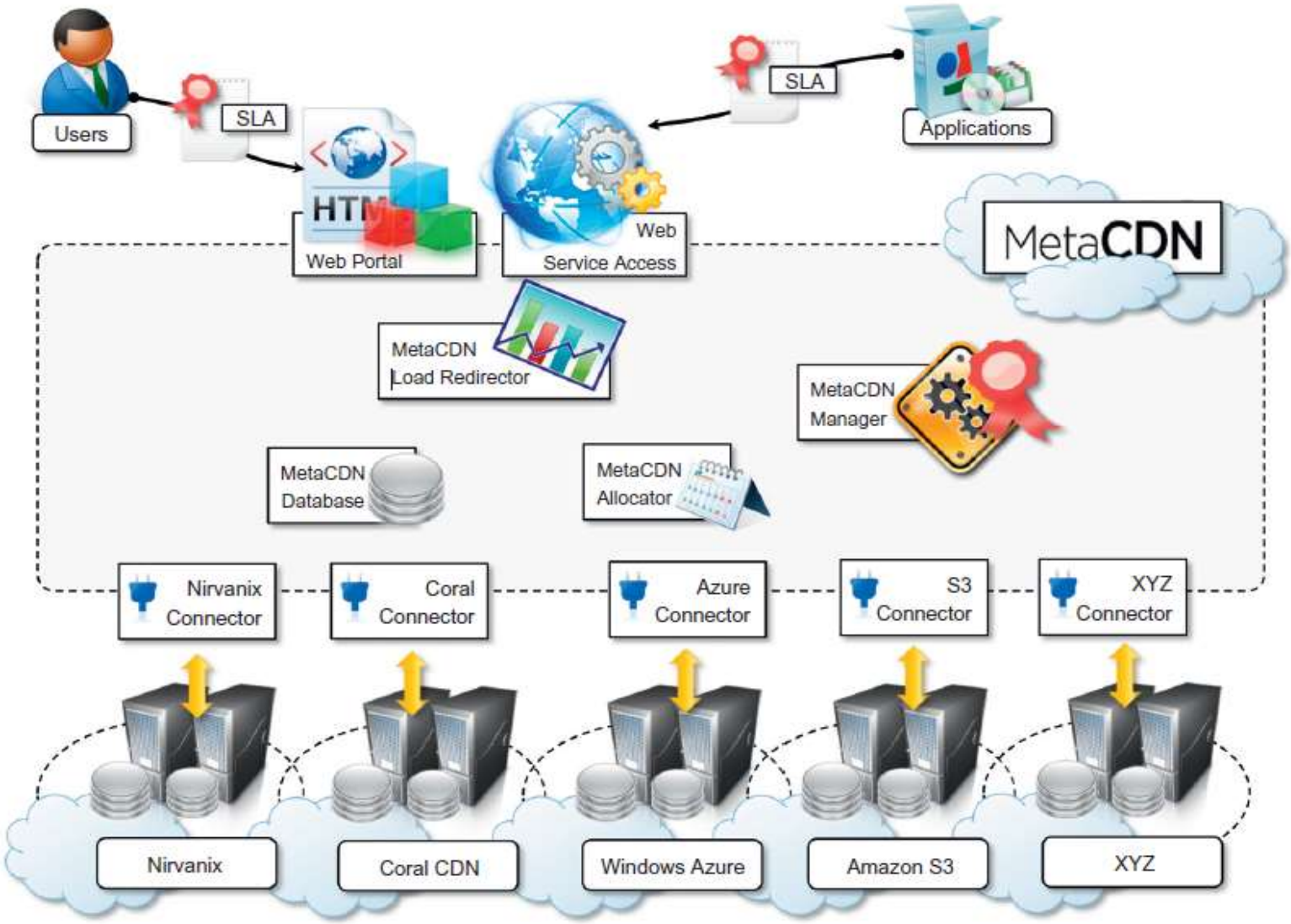


FIGURE 11.13
MetaCDN architecture.

MetaCDN Architecture

- The MetaCDN interface exposes its services through users and applications through the Web;
- **users interact with a portal,**
- while **applications take advantage of the Web services.**

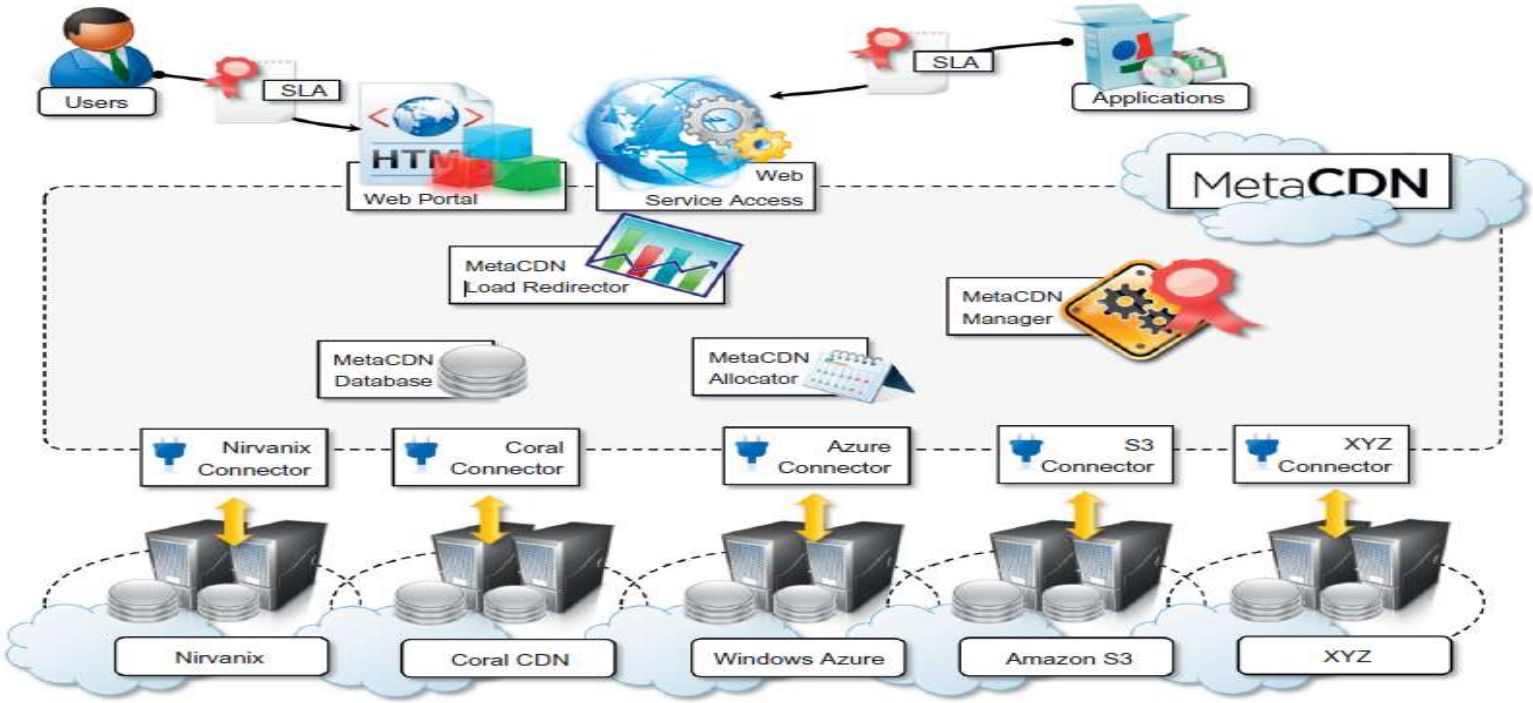


FIGURE 1.13
MetaCDN architecture.

MetaCDN Architecture

- The portal constitutes a more intuitive interface for users with basic requirements,
- The Web service provides access to the full capabilities of MetaCDN and allows for more complex and sophisticated deployment.

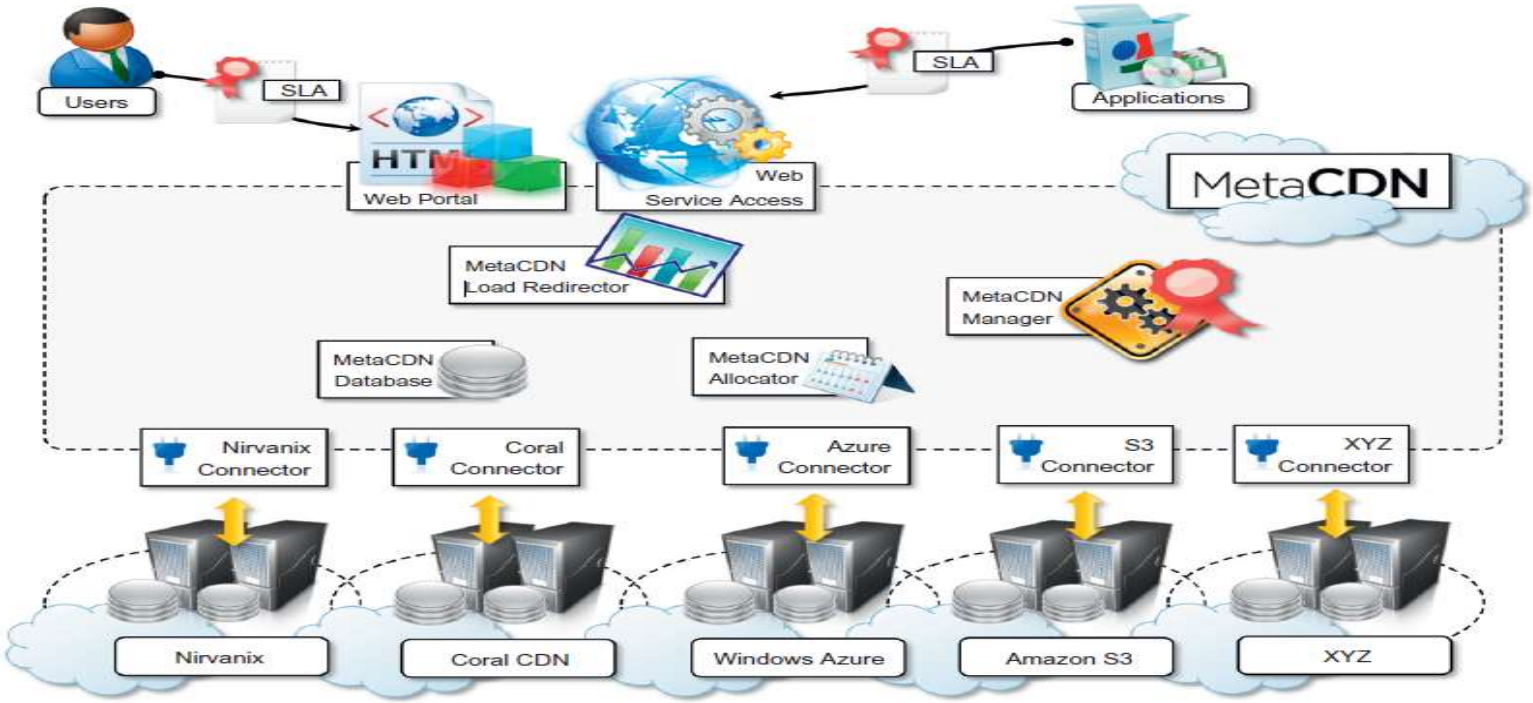


FIGURE 1.13
MetaCDN architecture.

MetaCDN Architecture

- The main operations of MetaCDN are the **creation of deployments over storage clouds and their management**

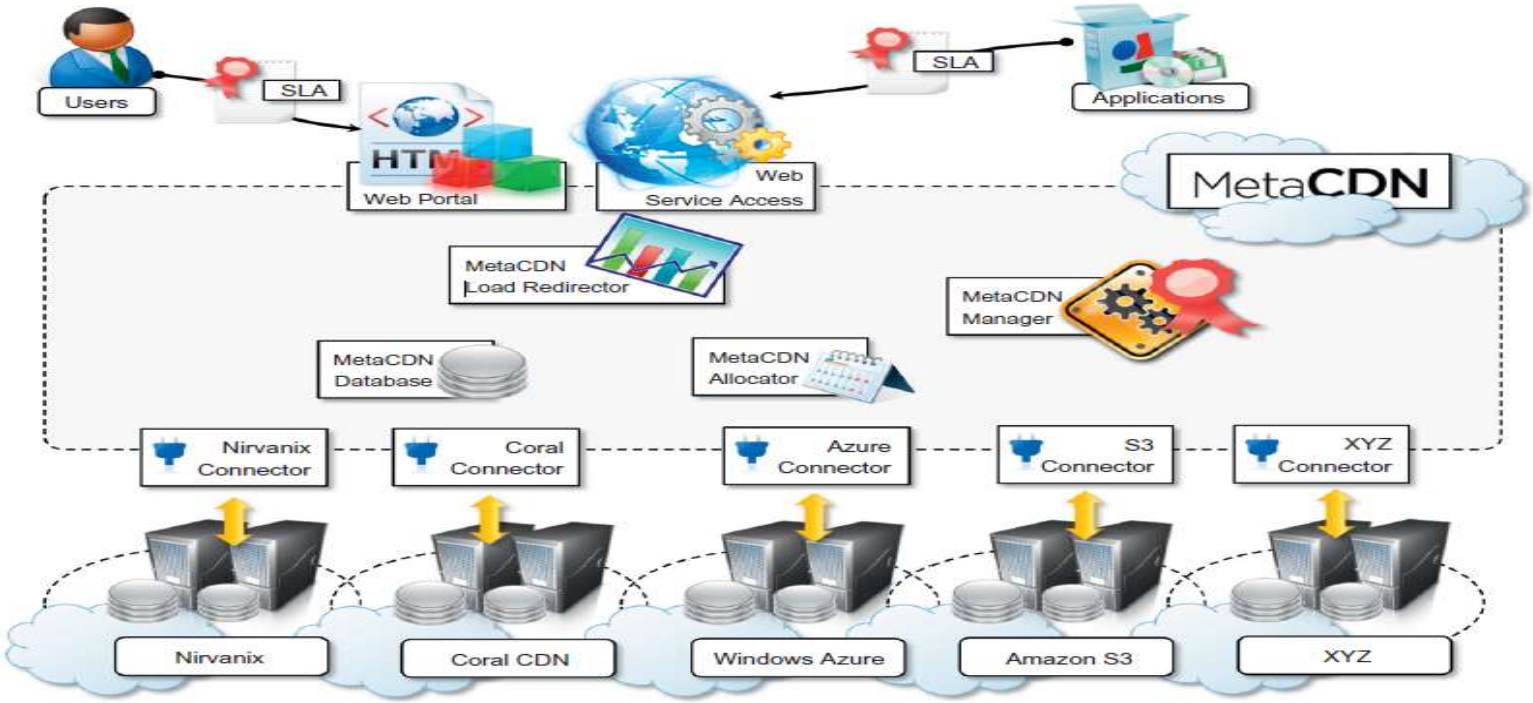
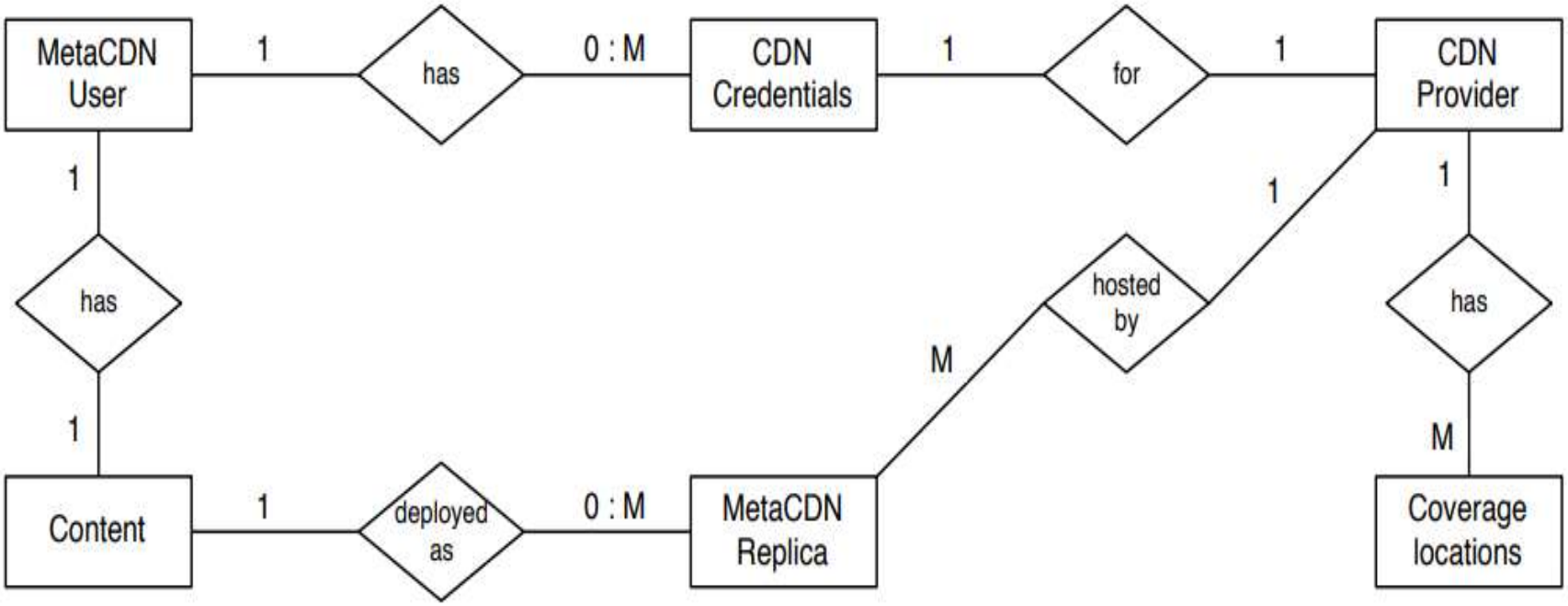


FIGURE 1.13
MetaCDN architecture.

MetaCDN Database



MetaCDN deployment options

In particular, four different deployment options can be selected:

- Coverage and performance-optimized deployment.
- Direct deployment.
- Cost-optimized deployment.
- QoS optimized deployment.

MetaCDN deployment options

In particular, four different deployment options can be selected:

- Coverage and performance-optimized deployment-
  - In this case MetaCDN **will deploy as many replicas as possible to all available locations**.
- Direct deployment-
  - In this case **MetaCDN allows the selection of the deployment regions for the content** and **will match the selected regions with the supported providers** serving those areas.

MetaCDN deployment options

- Cost-optimized deployment-
  - In this case MetaCDN deploys **as many replicas in the locations identified by the deployment request. The** available storage transfer allowance and **budget will be used to deploy the replicas and keep them active for as long as possible.**
- QoS optimized deployment-
  - **In this case MetaCDN selects the providers that can better match The QoS requirements attached to the deployment**, such as average response time and throughput from a particular location.

MetaCDN components

- Of particular importance are three components:
  - the MetaCDN Manager,
  - the MetaCDN QoS Monitor, and
  - the Load Redirector.

MetaCDN components

- The Manager is responsible for **ensuring that all the content deployments are meeting the expected QoS.**
- Manager is supported in this Activity by the Monitor, which constantly probes storage providers and **monitors data** transfers to **assess the performance of each provider.**

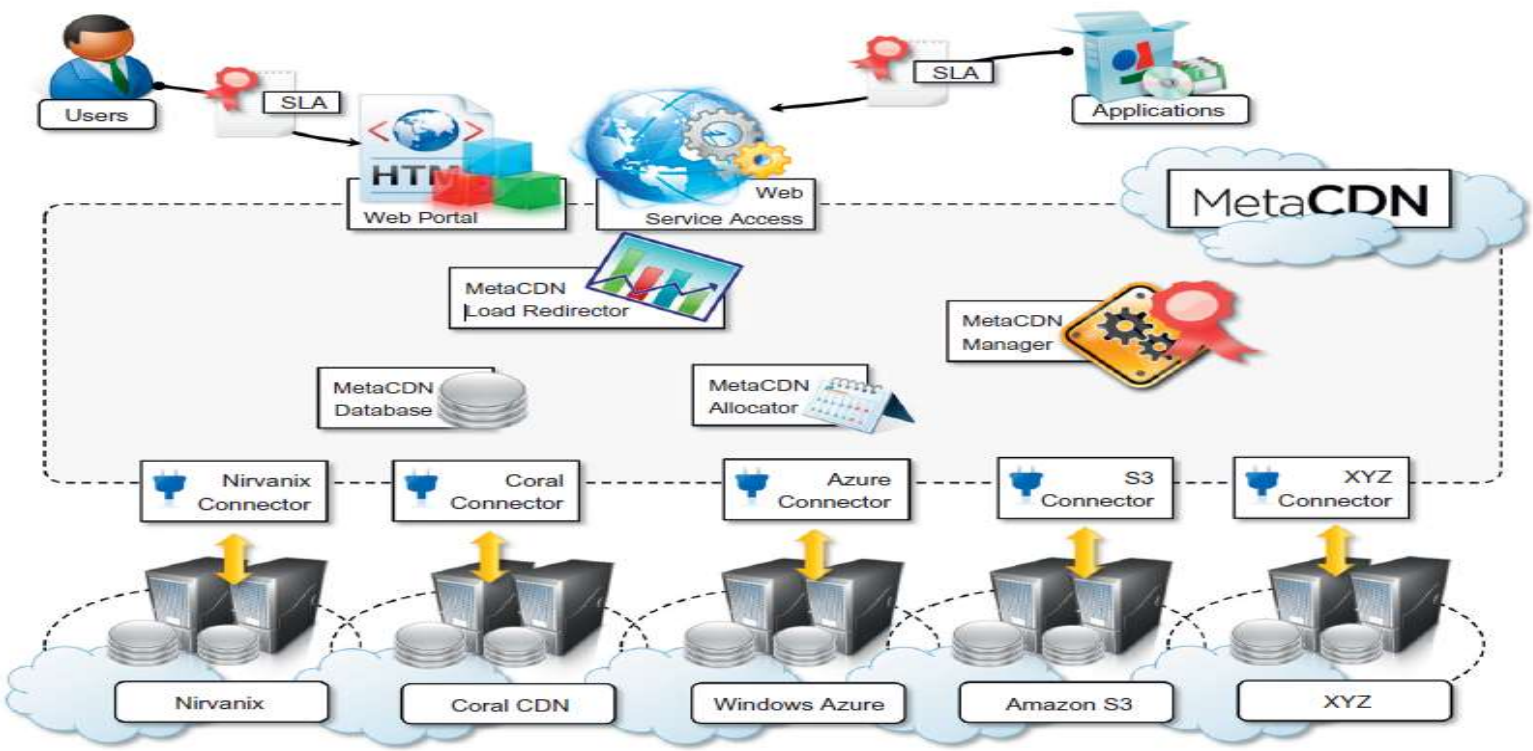


FIGURE 11.13
MetaCDN architecture.

MetaCDN components

- Content serving is controlled by the Load Redirector, Which is in-charge of **redirecting user content requests to the most suitable replica** given the condition of the systems and the options specified during the deployment.

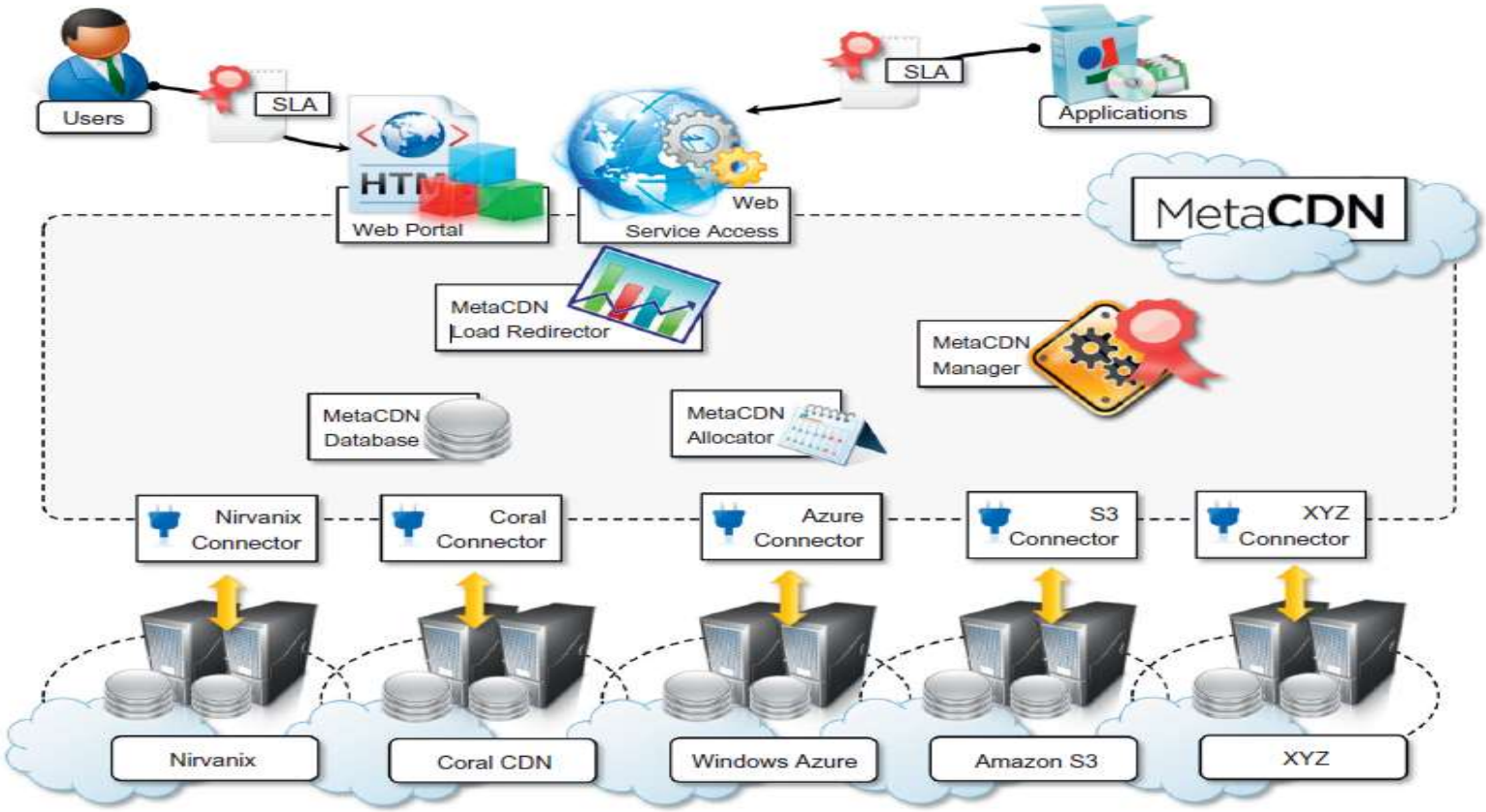


FIGURE 11.13
MetaCDN architecture.

MetaCDN components

- **Interactions with storage clouds are managed by means of connectors**, which **abstract away the different interfaces exposed by the providers and present a uniform interface** within the MetaCDN system.

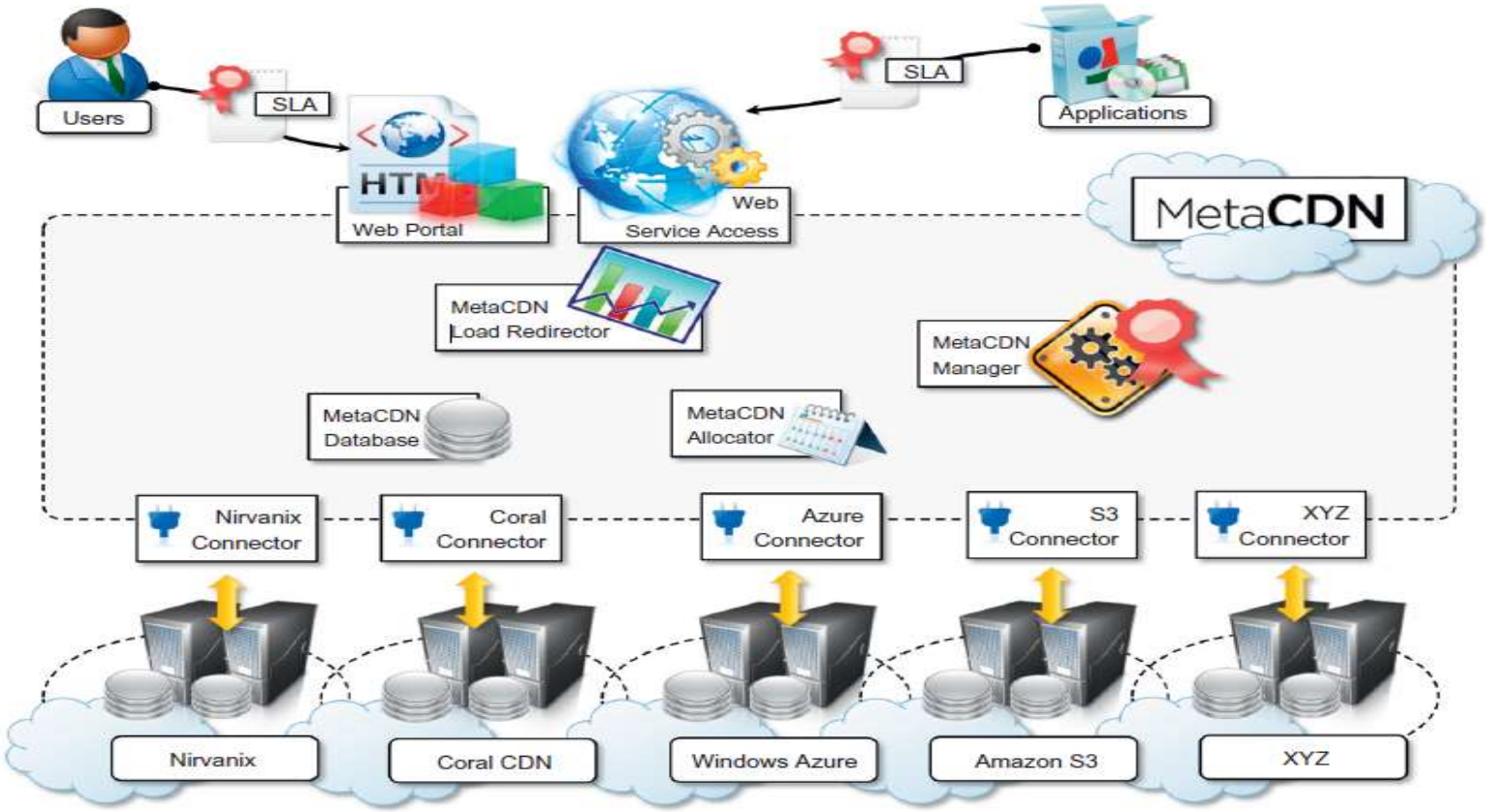
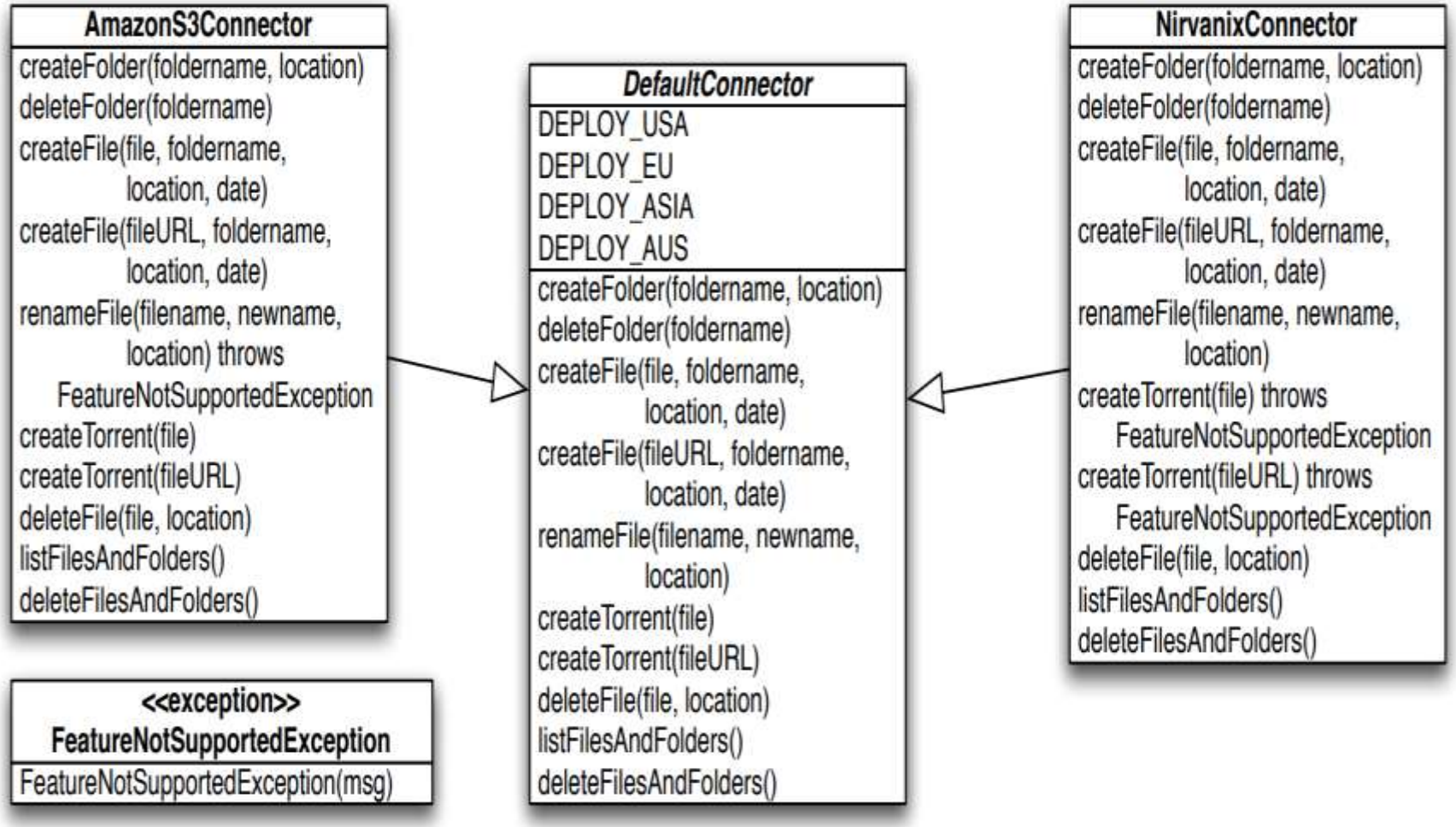
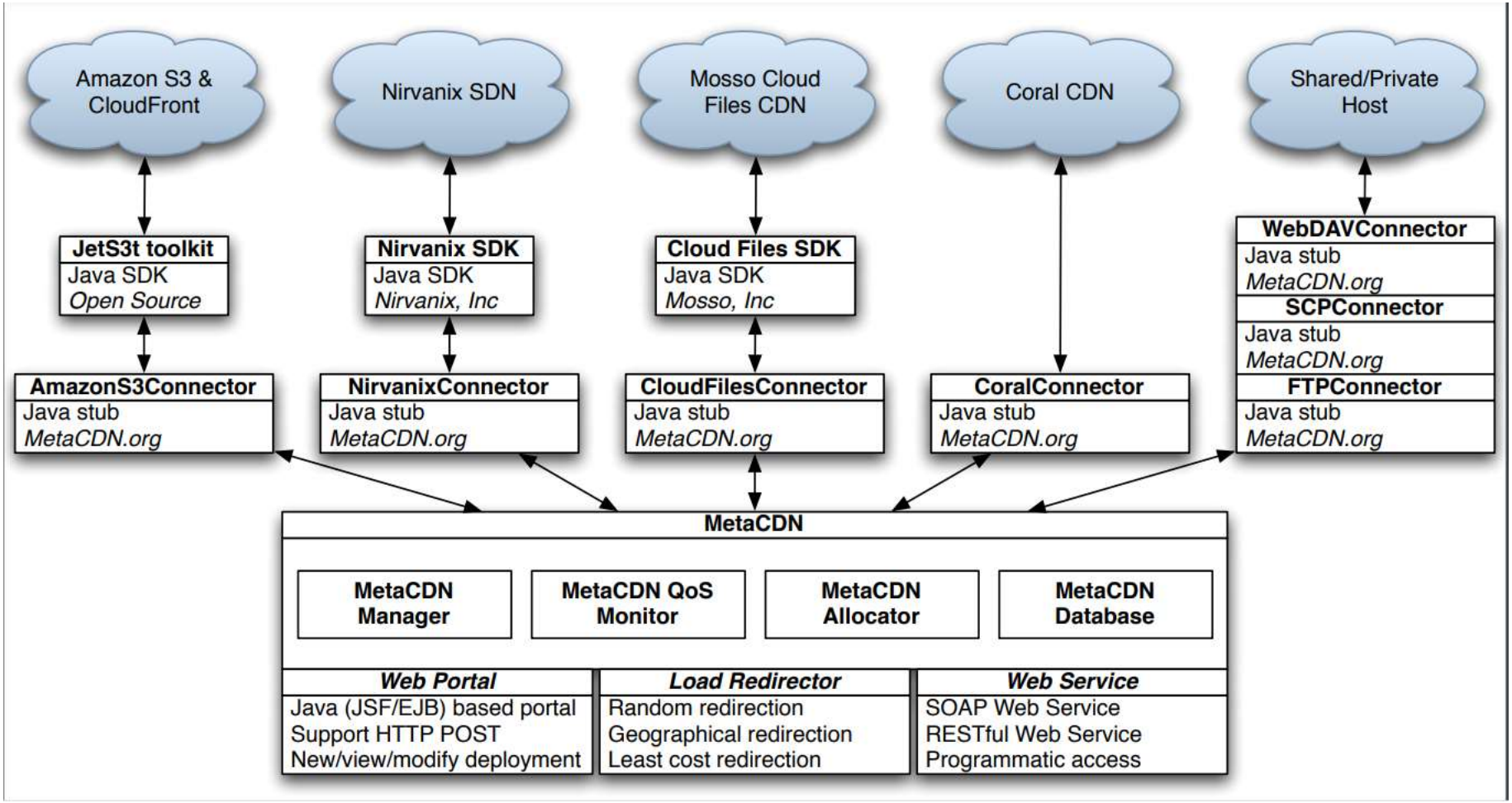


FIGURE 11.13
MetaCDN architecture.

How MetaCDN works



How MetaCDN works



SpotCloud

- SpotCloud is an **online portal that implements a virtual marketplace, where sellers and buyers can register and trade** cloud computing services.
- The platform is a **market place operating in the IaaS sector**.
- Buyers are looking for compute capacity that can meet the requirements of their applications, while sellers can make available their infrastructure to serve buyers

SpotCloud

- SpotCloud provides a comprehensive set of features that are expected for a virtual marketplace.
- Some of them include:
  - Detailed logging of all the buyers' transactions
  - Full metering, billing for any capacity
  - Full control over pricing and availability of capacity in the market
  - Management of quotas and utilization levels for providers
  - Federation management (many providers, many customers, but one platform)
  - Hybrid cloud support (internal and external resource management)
  - Full market administration and reporting
  - Applications and pre-build appliances directories

SpotCloud

- By acting as an **intermediary for trading compute and storage** between consumers and service providers, it provides the two parties with added value.

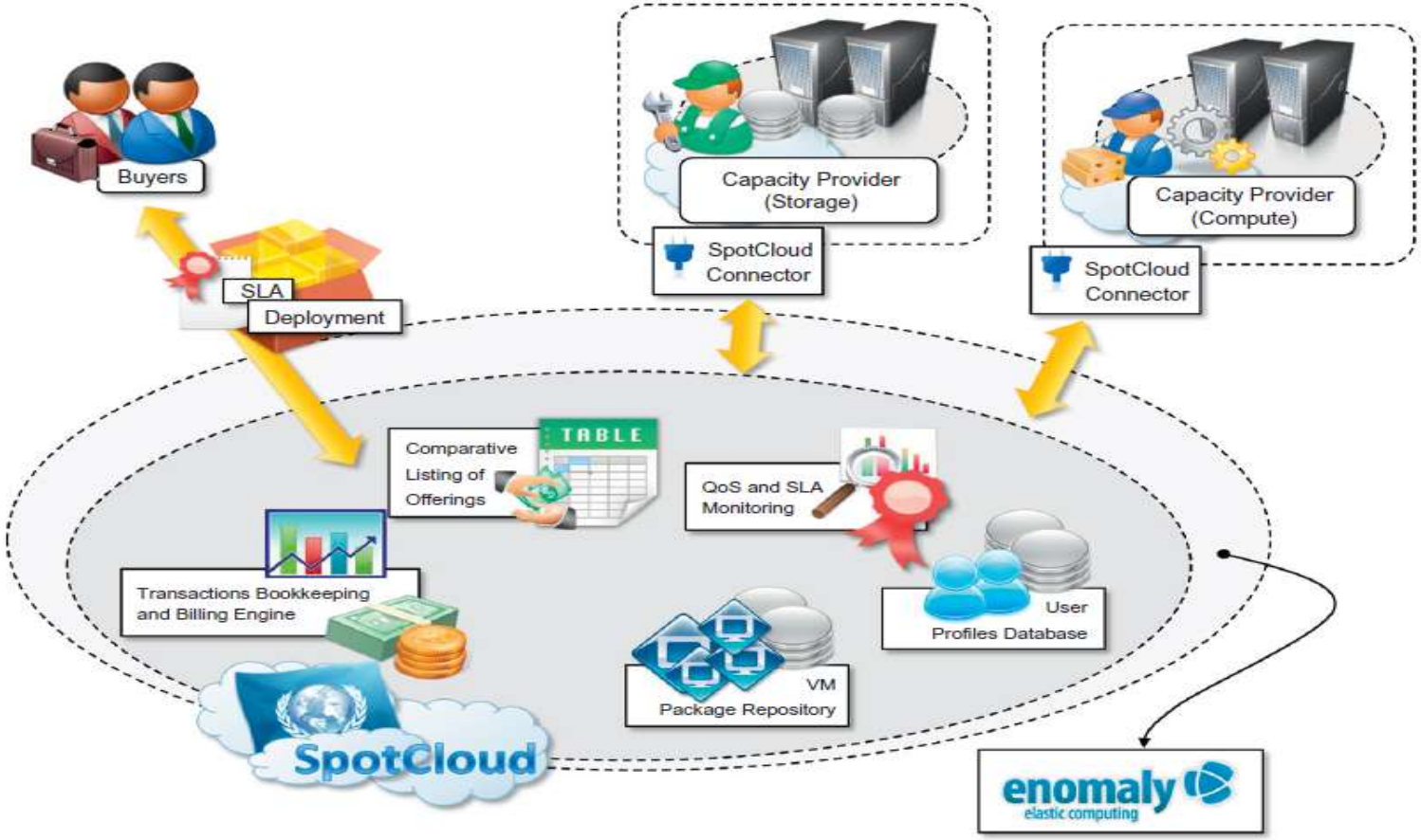


FIGURE 11.14
SpotCloud market architecture.

SpotCloud

- In addition, **it allows users with available computing capacity to easily turn themselves into service providers by deploying the runtime environment required by SpotCloud on their infrastructure**

SpotCloud

- **For service consumers, it acts as a market directory where they can browse and compare different IaaS service** offerings and select the most appropriate solution for them.
- **For service providers it constitutes an opportunity for advertising their offerings.**

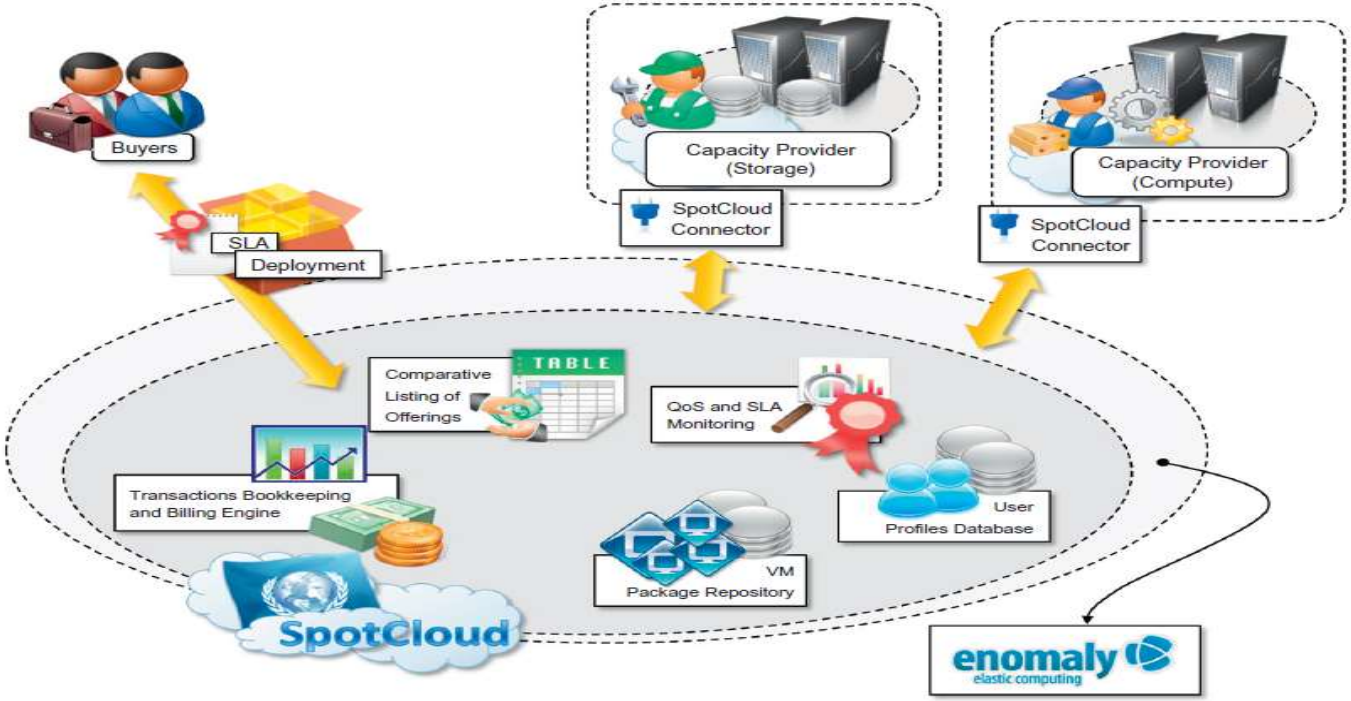


FIGURE 11.14
SpotCloud market architecture.

SpotCloud, the First Clearinghouse at Marketplace for Cloud Computing Services, Launched by Enomaly

??

Spotcloud

- **For cloud service providers**, the SpotCloud marketplace platform provides **an easy way to sell unused cloud capacity.**
- Cloud providers can use SpotCloud **to clear out unused capacity and sell computing inventory that would otherwise go unsold,** enabling **increased utilization and revenue, without undermining their standard pricing.**

SpotCloud

- SpotCloud is not only an enabler for IaaS providers and resellers,
- **Also includes a complete bookkeeping of the transactions associated with the use of resources.**

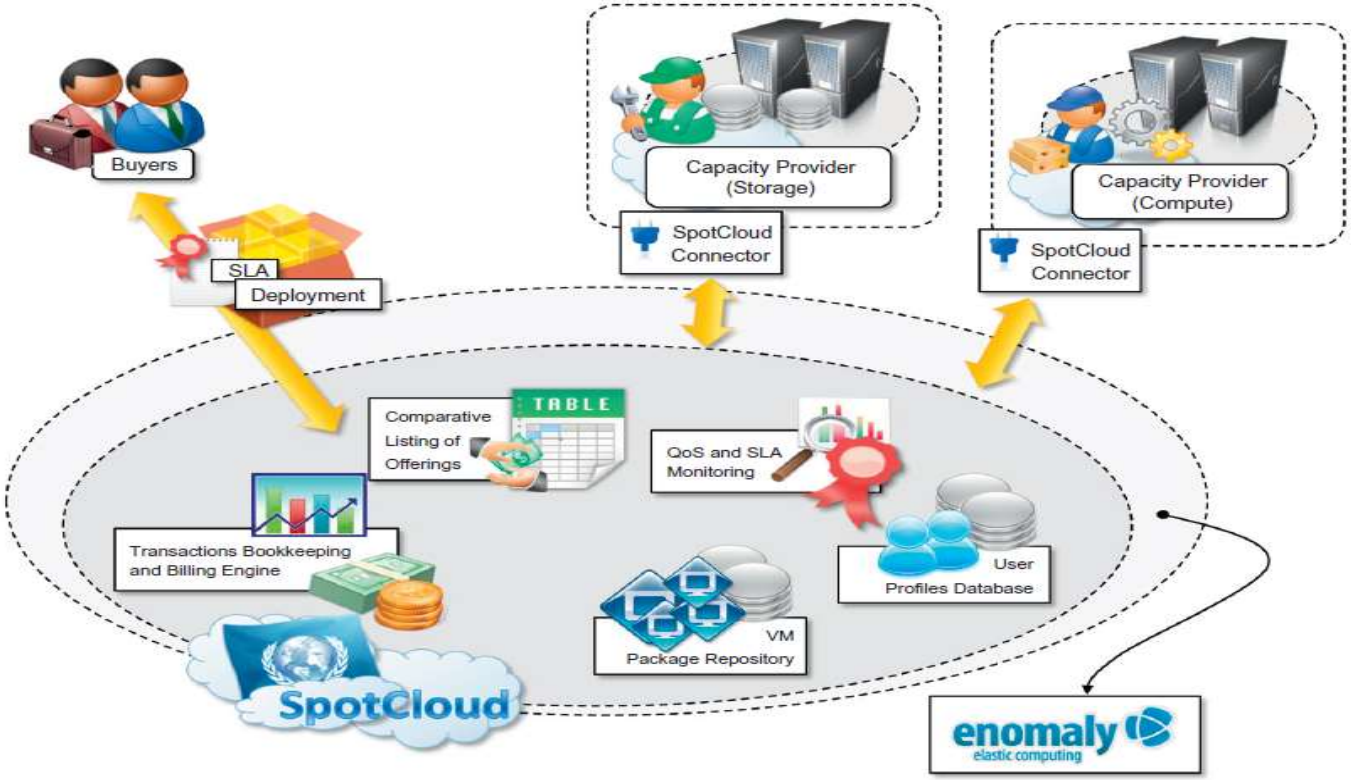


FIGURE 11.14
SpotCloud market architecture.

SpotCloud

- **Users deposit credit on their SpotCloud account and capacity sellers are paid following the usual pay- per-use model.**
- SpotCloud **retains a percentage of the amount billed to the user.**

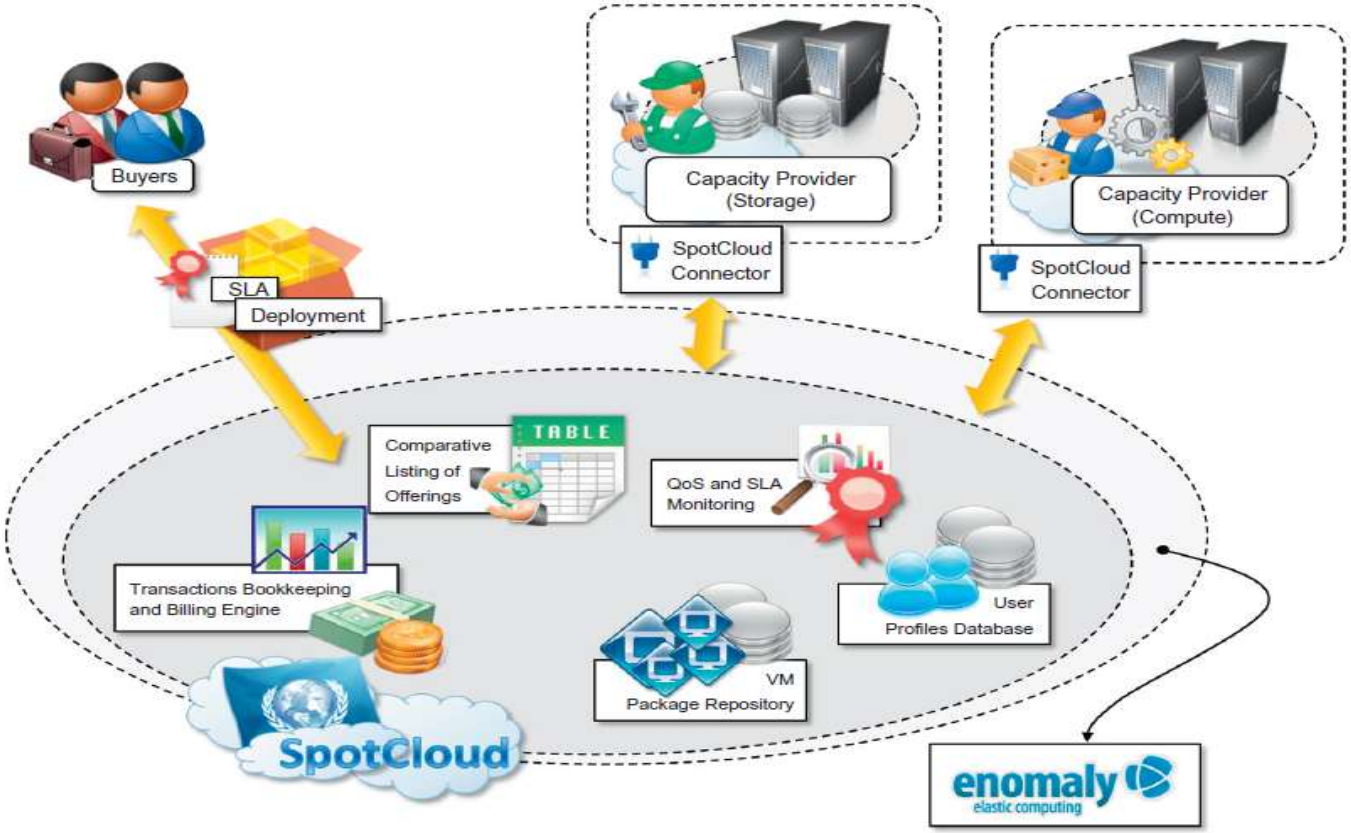


FIGURE 11.14
SpotCloud market architecture.

SpotCloud

- Moreover, by leveraging a **uniform runtime environment and virtual machine management layer, it provides users with a vendor lock-in-free solution,** which might be strategic for specific applications.

Federated Clouds/Intercloud

- These are enablers for MOCC since they provide **means for interoperation among different cloud providers.**
- **Building aggregations of clouds that belong to different administrative domains.**

Federated Clouds

Federation

- **Creation of an organization that supersedes the decisional and administrative power** of the single entities and that acts as a whole.

- Within a cloud computing context, **the word federation implies that there are agreements between the various cloud providers**, allowing them to leverage each other's services in a privileged manner.

Federated Clouds

- Definition:-

“Cloud federation manages consistency and access controls when two or more independent geographically distinct Clouds share either authentication, files, computing resources, command and control or access to storage resources”

-By Reuven Cohen, founder and CTO of Enomaly Inc.

Technologies for cloud federations

- **Even though the concept of cloud federation or the InterCloud is still immature**, there are some supporting technologies that enable the deployment of interoperable clouds:-
- RESERVOIR

RESERVOIR

- **Resources and Services Virtualization Without Barriers, or RESERVOIR**
- A European research project focused on developing an architecture that supports **providers of cloud infrastructures to dynamically partner with each other to extend their capabilities while preserving their administrative autonomy.**

RESERVOIR Architecture

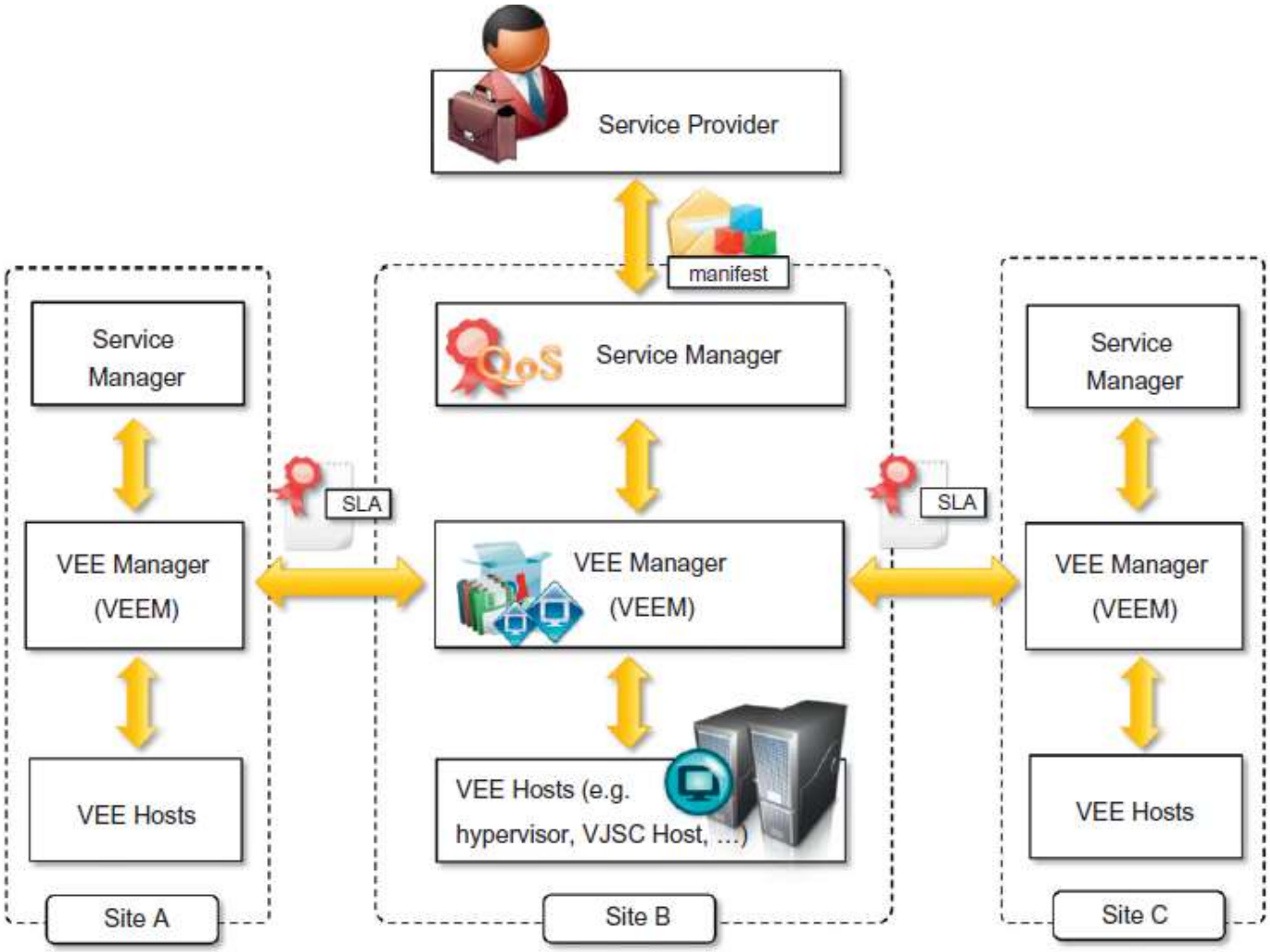


FIGURE 11.11

RESERVOIR architecture.

RESERVOIR Architecture

Service Manager –

- **Constitutes the front-end used by service providers to submit service** manifests, negotiate pricing, and monitor applications.
- **This component deploys and provisions VEEs** according to the service manifest and **monitors and enforces SLA compliance by controlling the capacity** of a service application.

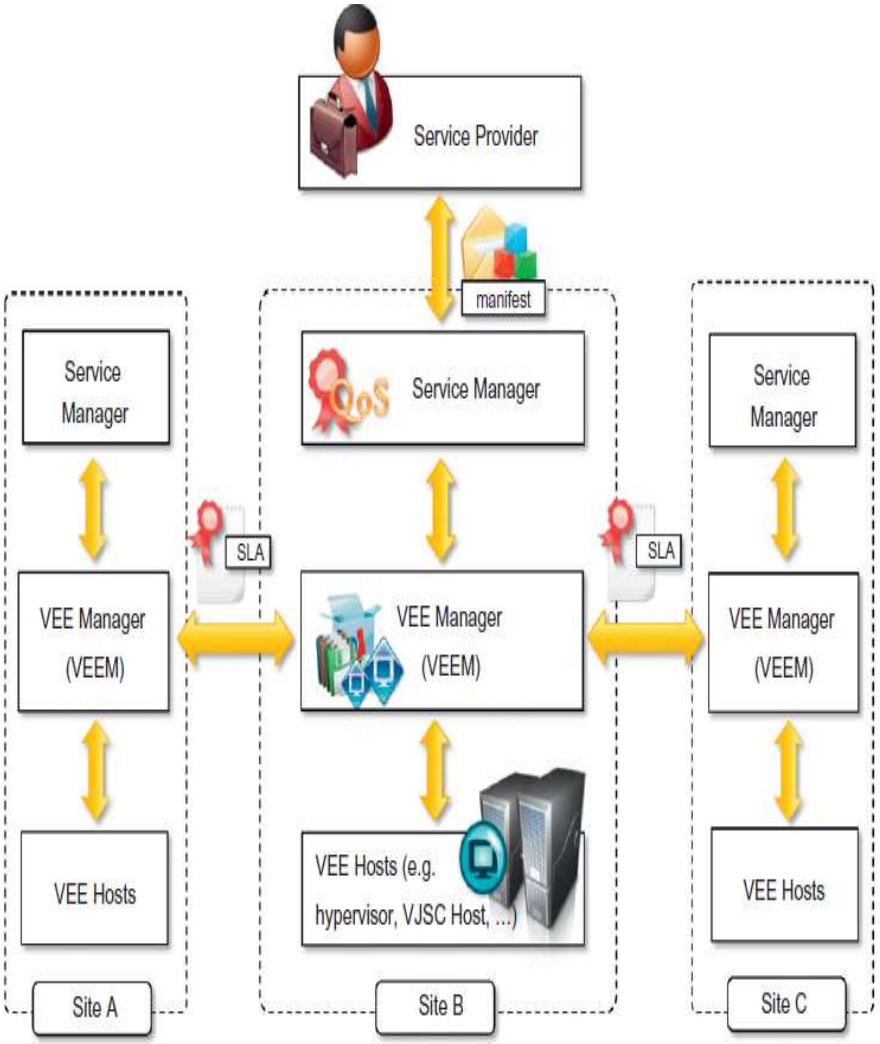


FIGURE 11.11

RESERVOIR architecture.

RESERVOIR Architecture

Virtual Execution Environment (VEE) Manager –

- Core of the RESERVOIR middleware
- **is responsible for the optimal placement of VEEs into VEE hosts according to the constraints expressed by the Service Manager.**

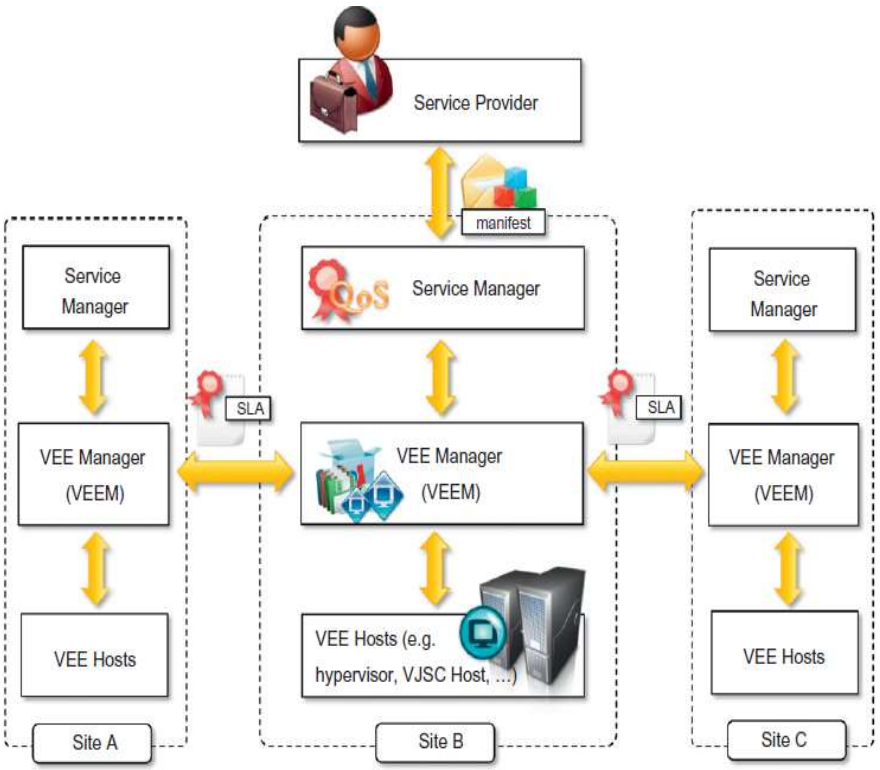


FIGURE 11.11

RESERVOIR architecture.

RESERVOIR Architecture

Virtual Execution Environment (VEE) Manager –

- Moreover, the VEE Manager **also interacts with VEE Managers in other sites to provision additional instances for the execution of service applications or move VEEs to other sites in case of overload.**
- This component **realizes the cloud federation.**

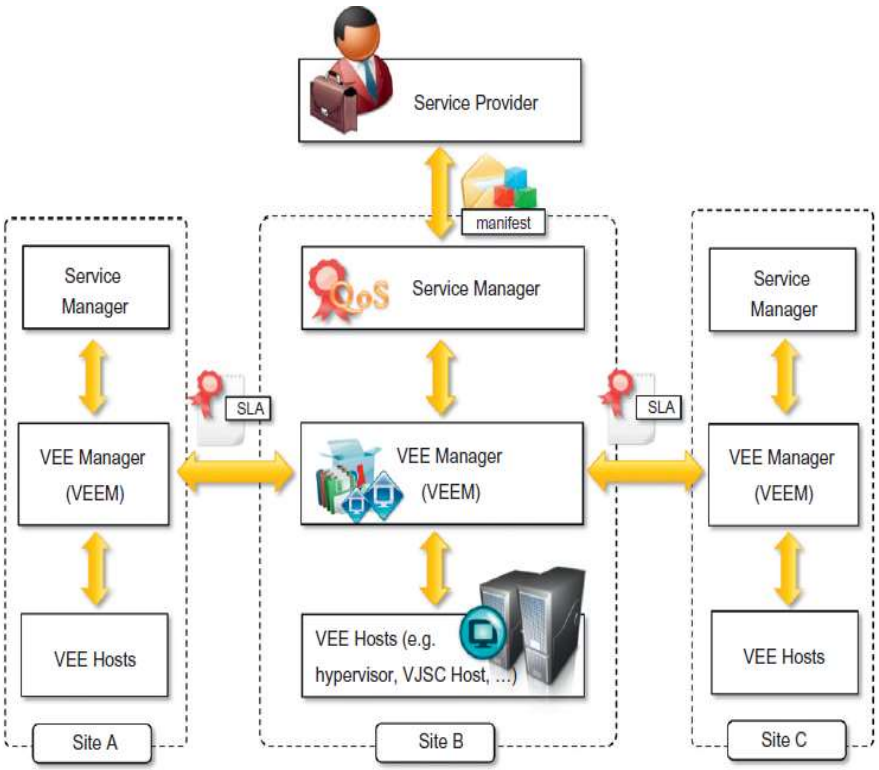


FIGURE 11.11

RESERVOIR architecture.

RESERVOIR Architecture

VEE Host(VEEH) –

- Interacts with the VEE Manager to **put into practice the IT management decisions regarding heterogeneous sets of virtualization platforms.**
- Ensures **networking among VEEs that belong to the same application**

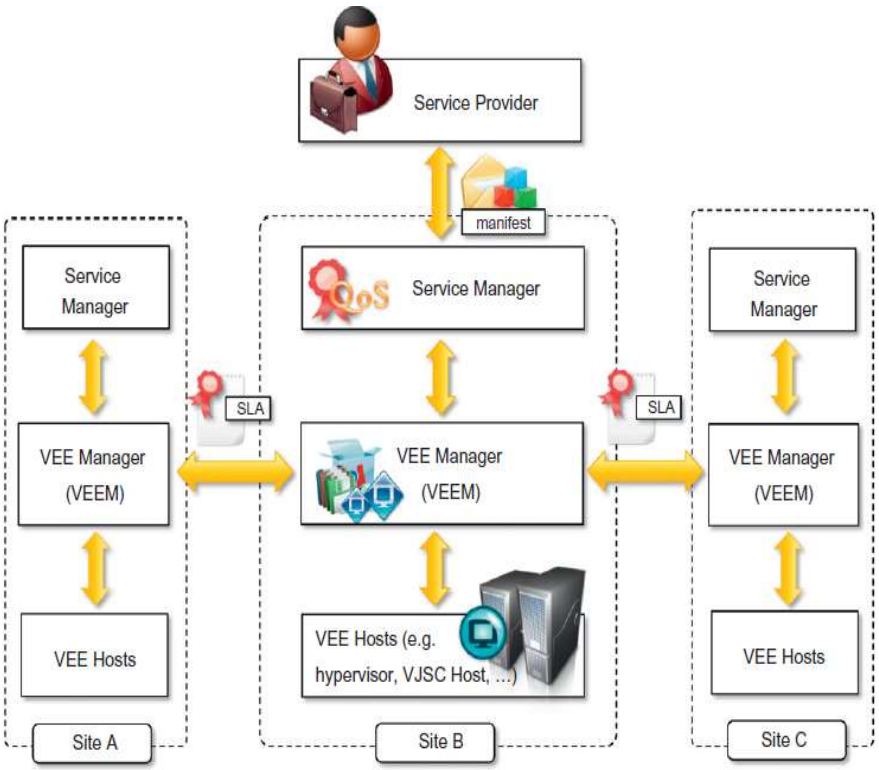


FIGURE 11.11
RESERVOIR architecture.

RESERVOIR Architecture

VEE Host(VEEH) –

- The VEEH **encapsulates all platform-specific management** that is required to expose the used virtualization technology **through a standardized interface to the VEE Manager.**

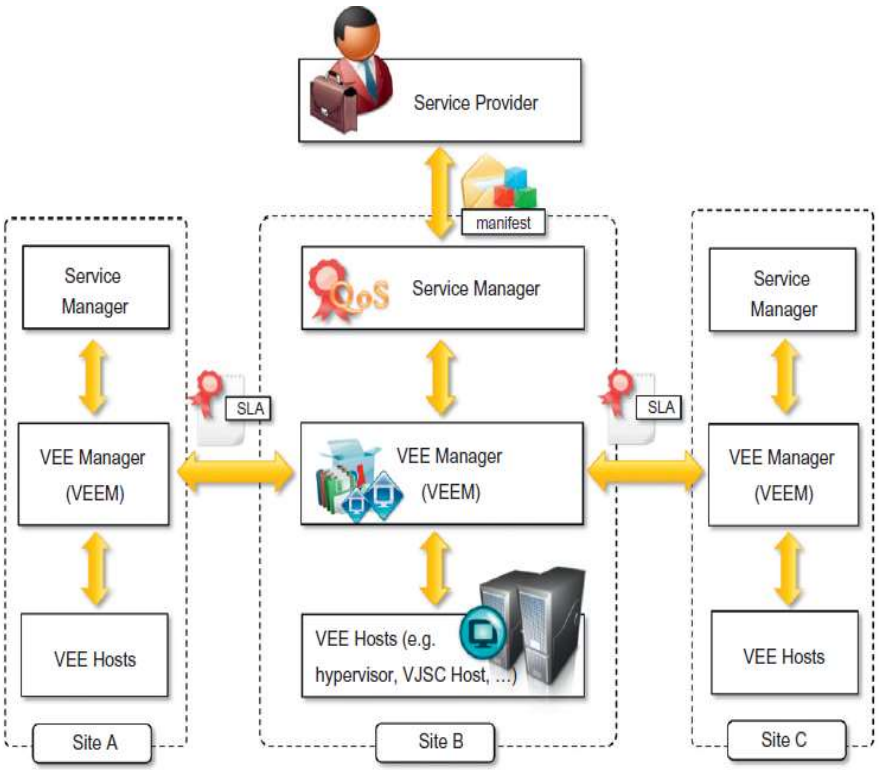


FIGURE 11.11

RESERVOIR architecture.

RESERVOIR Cloud deployment

- The framework defines an infrastructure overlay that spans multiple administrative domains and different geographic locations.
- **Each site runs the RESERVOIR software stack**

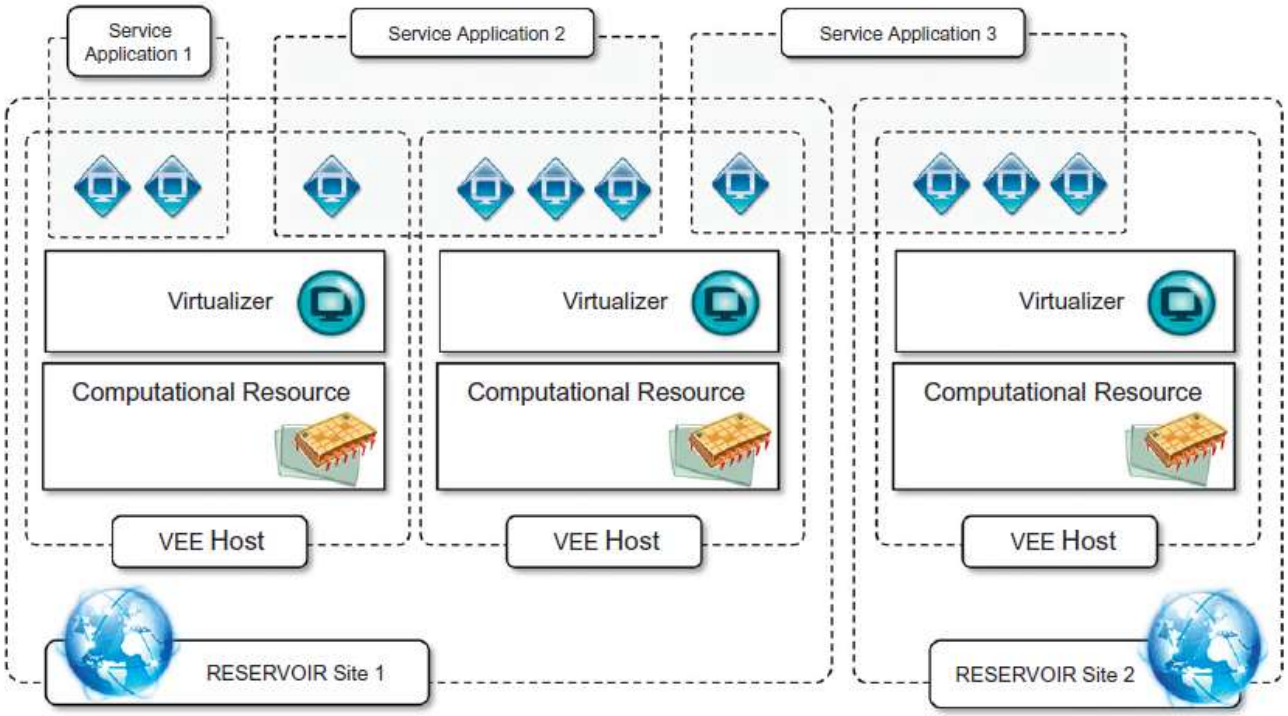


FIGURE 11.10

RESERVOIR cloud deployment.

RESERVOIR Cloud deployment

- **Infrastructure providers operate RESERVOIR sites**
- **Service providers define service applications**

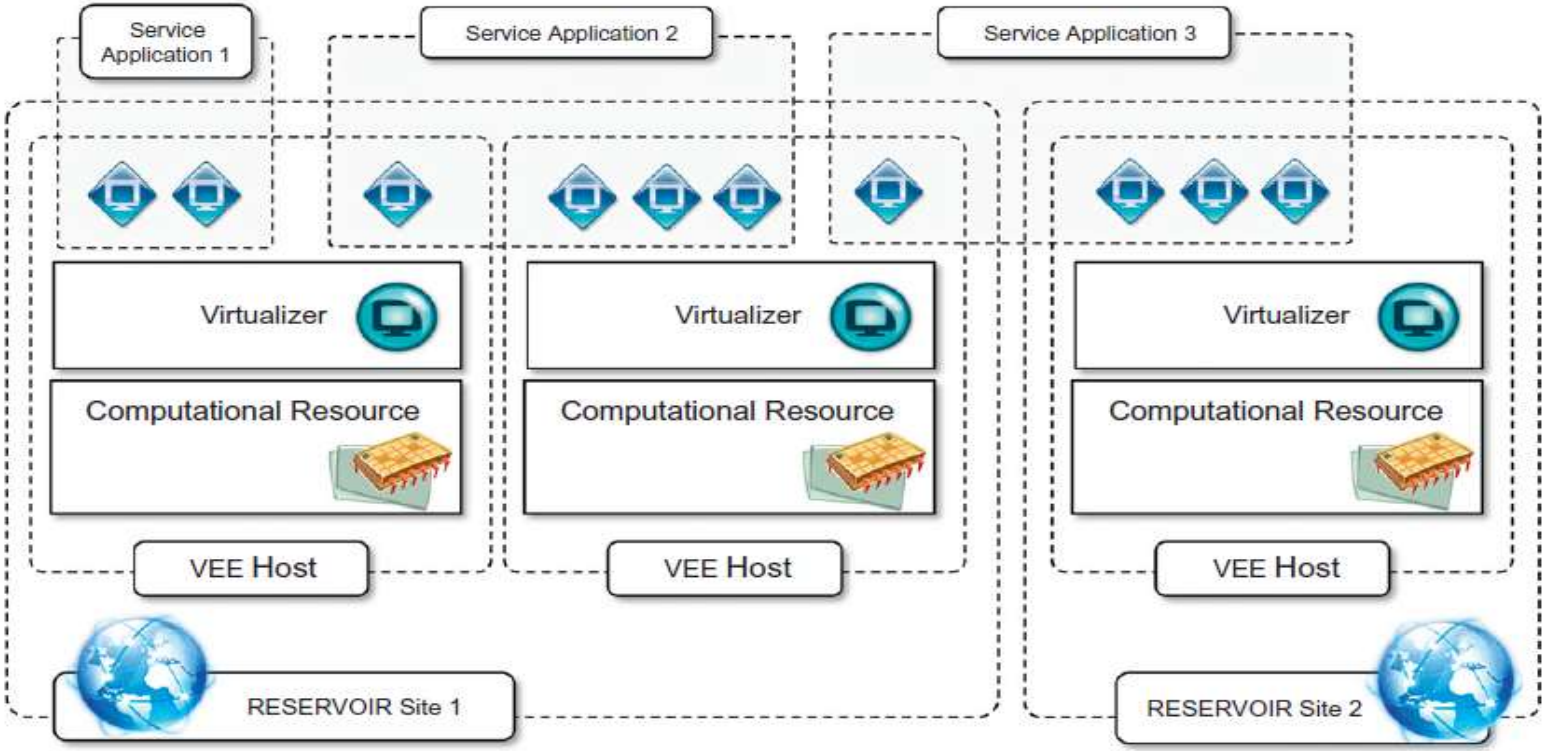


FIGURE 11.10

RESERVOIR cloud deployment.

RESERVOIR Cloud deployment

- In the first case, the **service provider conducts sizing to identify the appropriate number of components to be required for a given workload condition.**
- The specification is obtained **by means of minimal service configuration and a set of elasticity rules that are used by RESERVOIR**

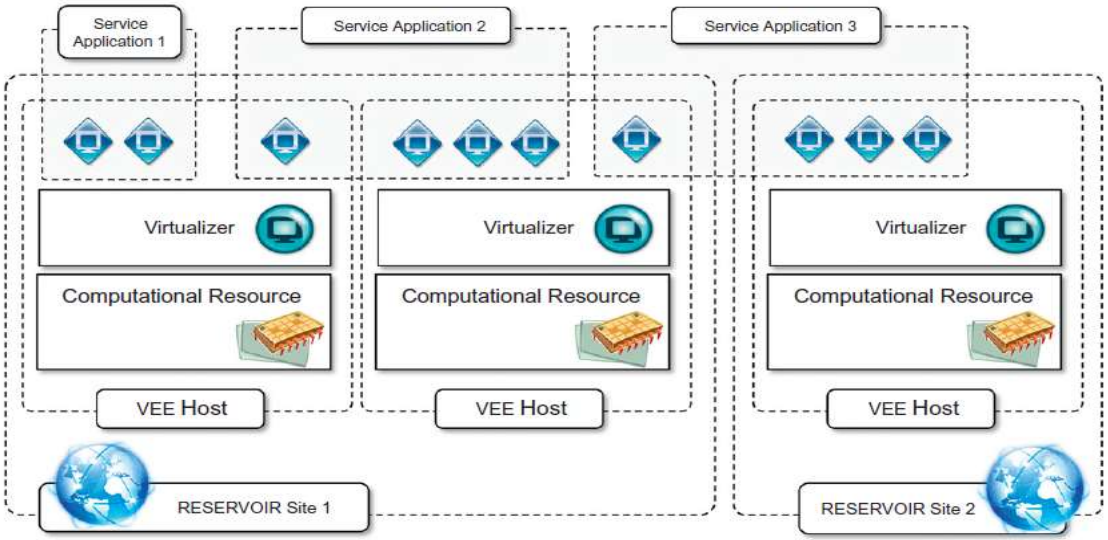


FIGURE 11.10
RESERVOIR cloud deployment.

RESERVOIR Cloud deployment

- In the second case, the service provider provides **neither a minimal service configuration nor elasticity rules.**
- **The sizing is automatically made by the RESERVOIR middleware, which tries to minimize overprovisioning.**

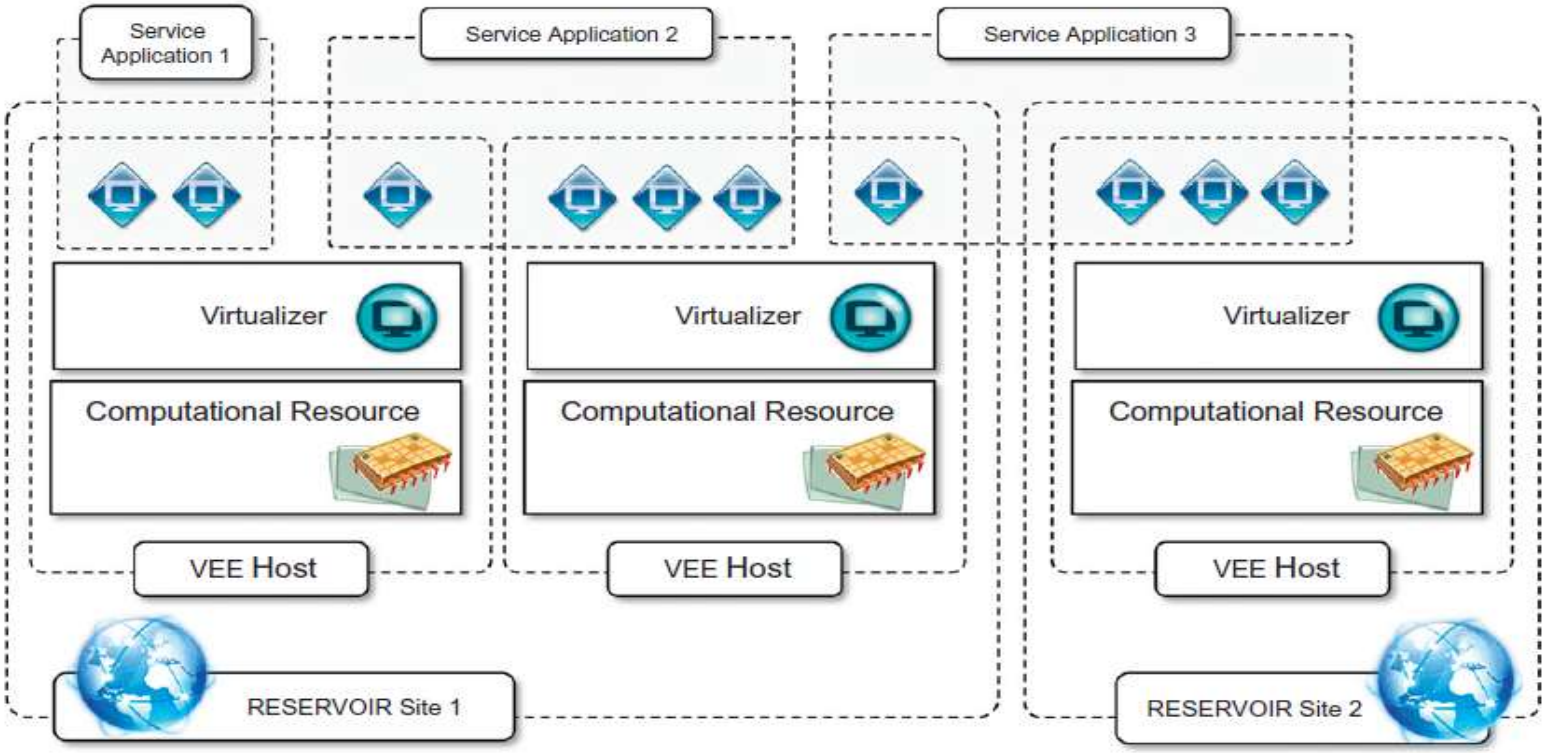


FIGURE 11.10

RESERVOIR cloud deployment.

Intercloud

- InterCloud is a term that is often **used interchangeably to express the concept of Cloud federation.**
- **It was introduced by Cisco** for expressing **a composition of clouds that are interconnected by means of open standards** to provide a universal environment that leverages cloud computing services.

Intercloud

- By mimicking the Internet term, often referred as the "network of networks,"
- **InterCloud represents a "Cloud of Clouds"**

Intercloud

- Intercloud Expresses the same concept of federating together clouds
- In many cases acceptable,
- Some practitioners and experts like Ellen Rubin,founder and VP of Products at CloudSwitch— **prefer to give different connotations to the two terms:**

Intercloud

- InterCloud refers mostly to a global vision in which interoperability among different cloud providers is **governed by standards, thus creating an open platform where applications can shift workloads and freely compose services from different sources.**
- On the other hand, the concept of a cloud federation is more general and **includes ad hoc aggregations between cloud providers on the basis of private agreements and proprietary interfaces.**

InterCloud Architecture

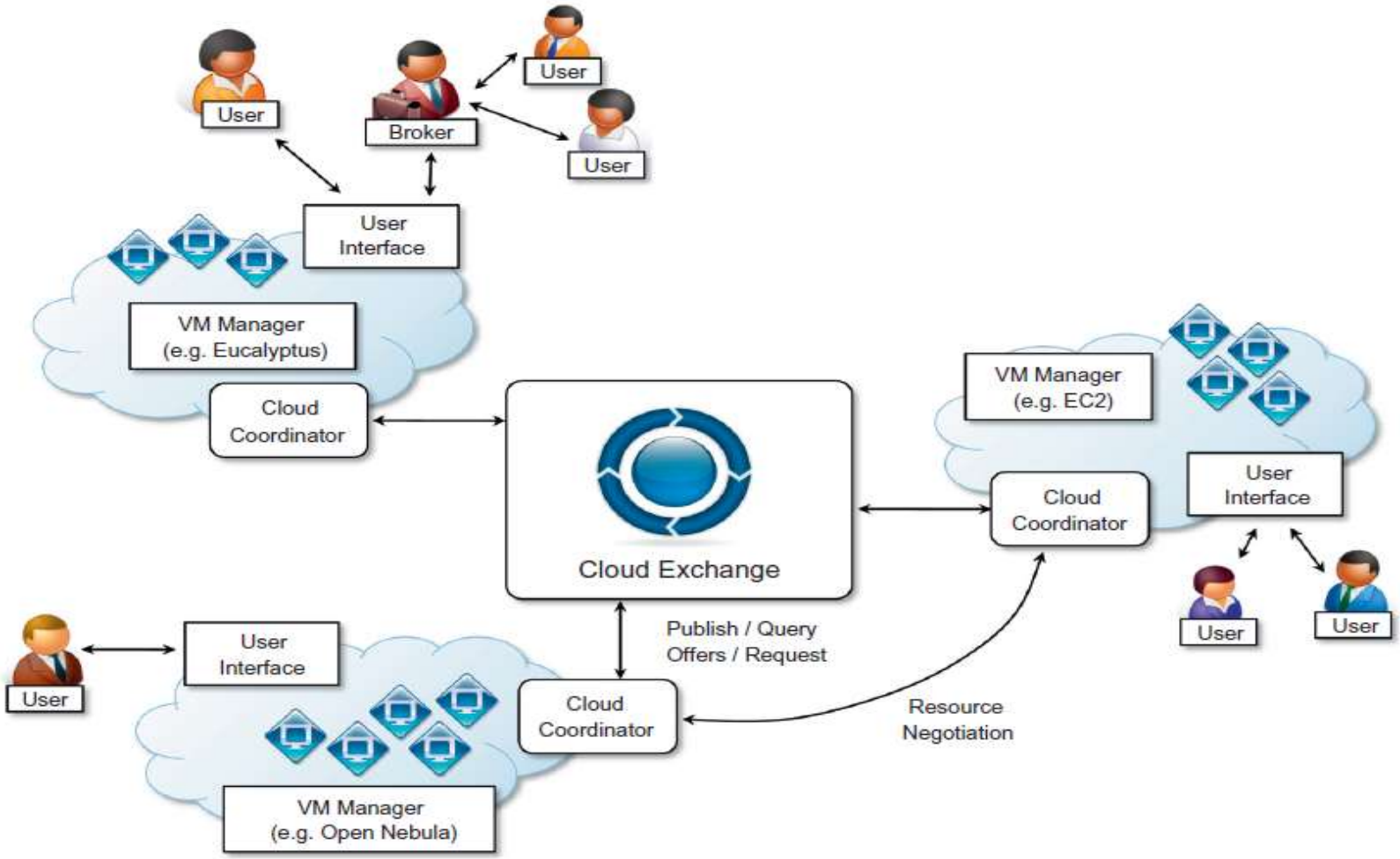


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

- InterCloud is a service-oriented architectural framework for cloud federation that supports utility-driven interconnection of clouds

InterCloud Architecture

The InterCloud model comprises two main elements:

- CloudExchange
- CloudCoordinator

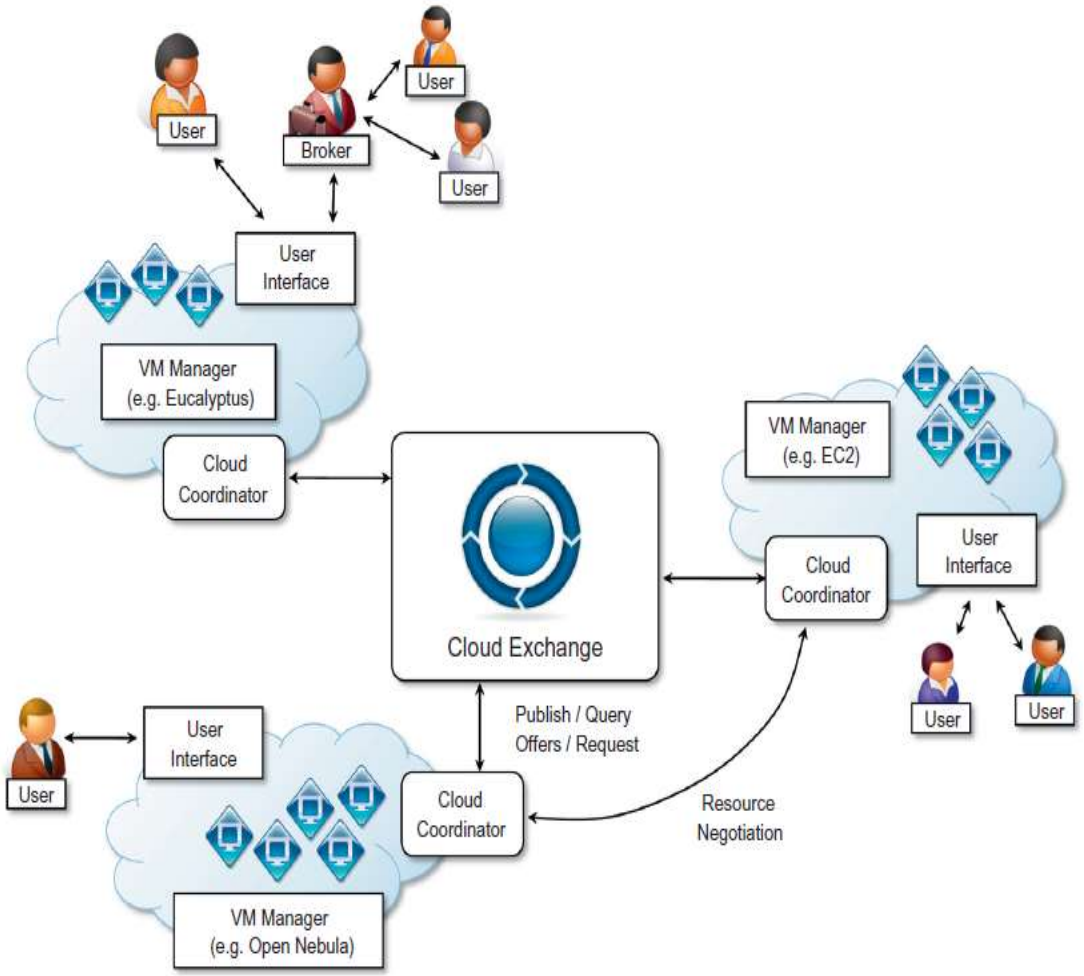


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

CloudExchange

- In the former case, **CloudExchange acts as a directory service** for the federation.
- In the latter case, **it runs the auction.**

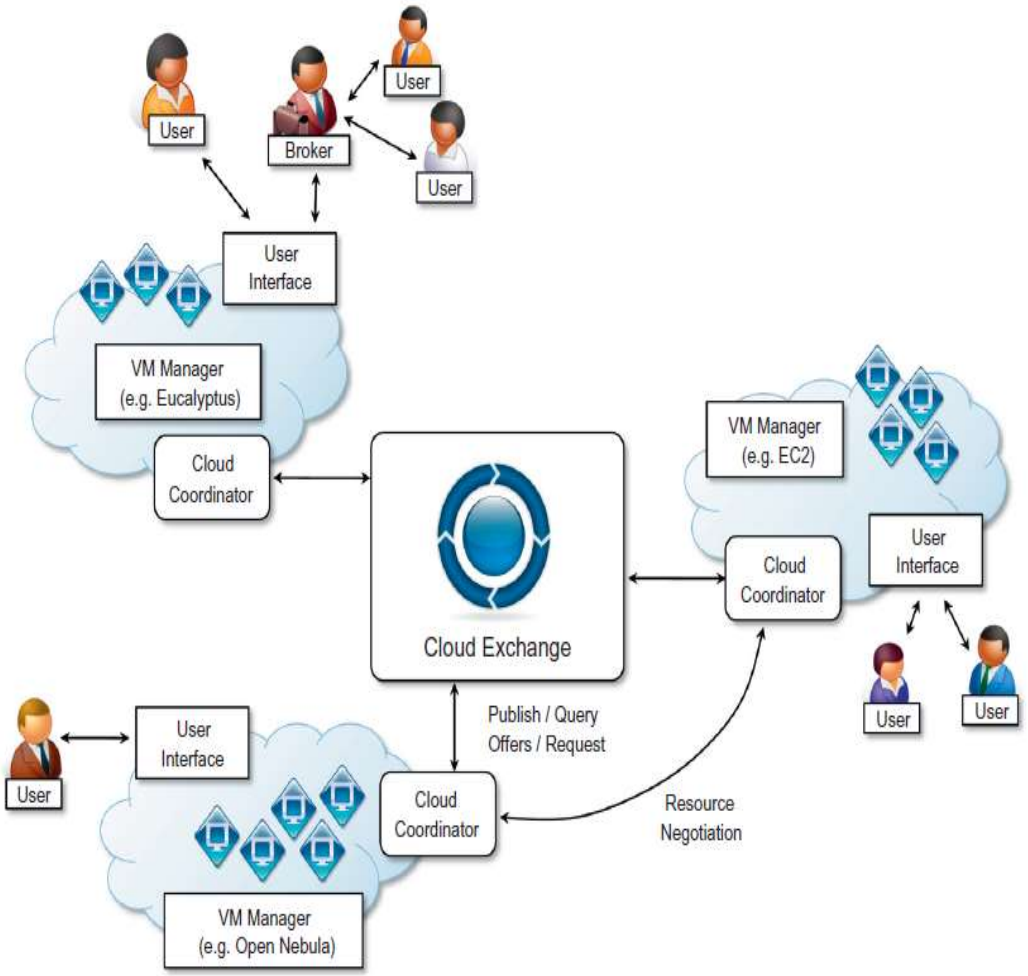


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

CloudExchange

- This is the **market-making component of the architecture.**
- It offers **services that allow providers to find each other** in order to directly trade cloud assets, as well as **allowing parties to register and run auctions.**

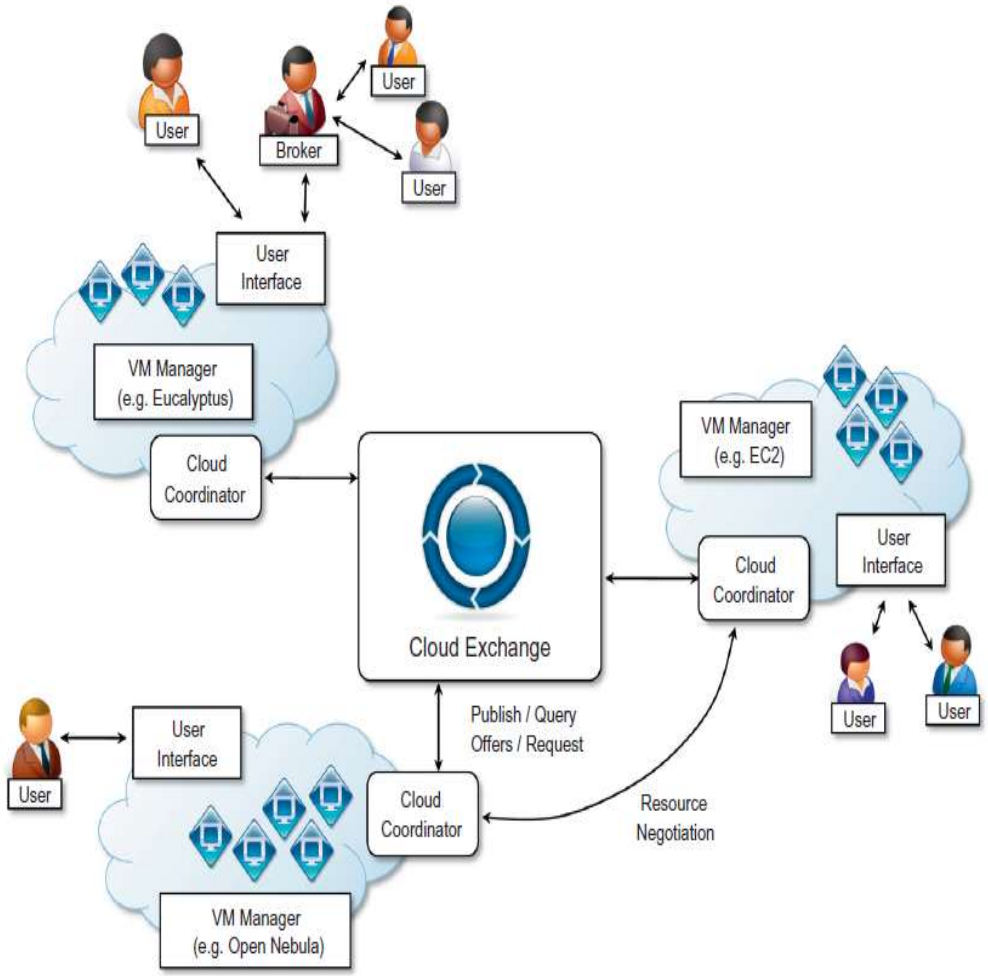


FIGURE 11.12
InterCloud architecture.

InterCloud Architecture

CloudCoordinator

CloudCoordinator has

- **Front-end components** (i.e., elements that interact with the federation)
- **Back-end components** (i.e., components that interact with the associated datacenter).

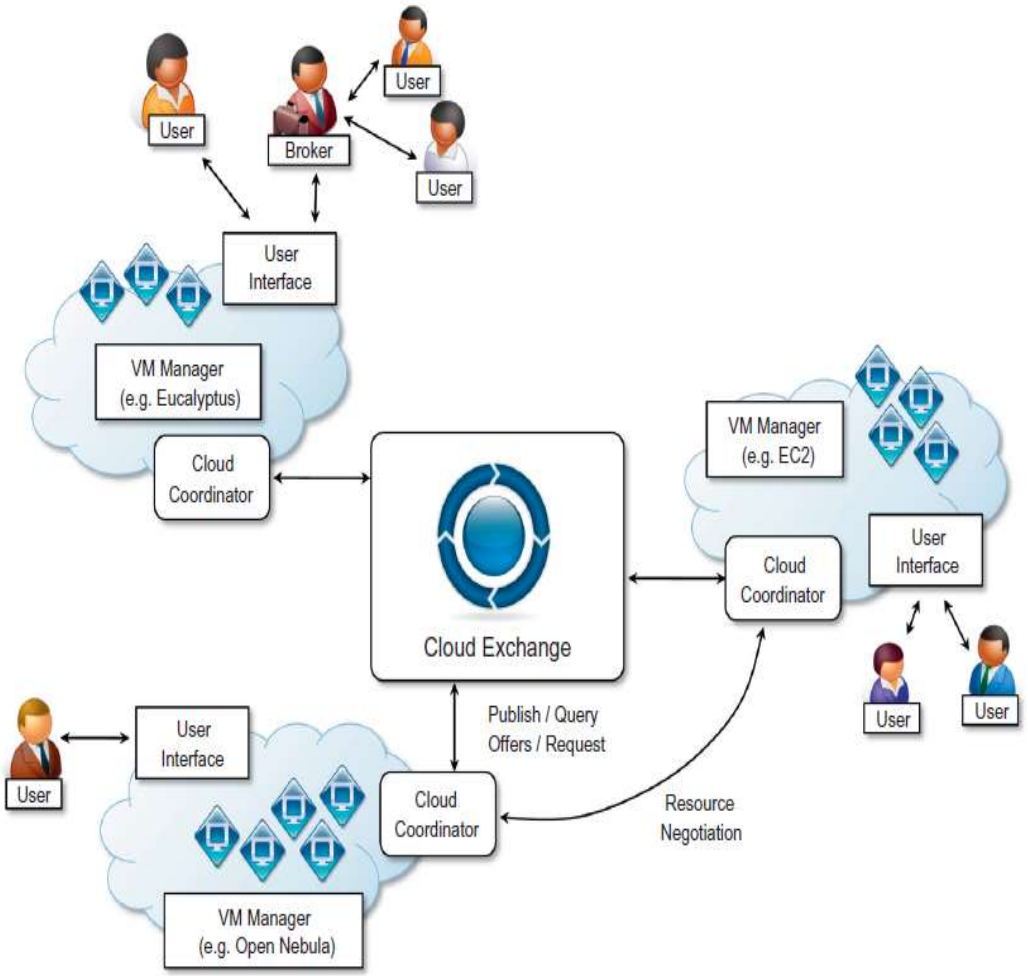


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

CloudCoordinator

Front-end components **interact with the CloudExchange and with other coordinators.**

- **Allows datacenters to announce their offers and requirements**

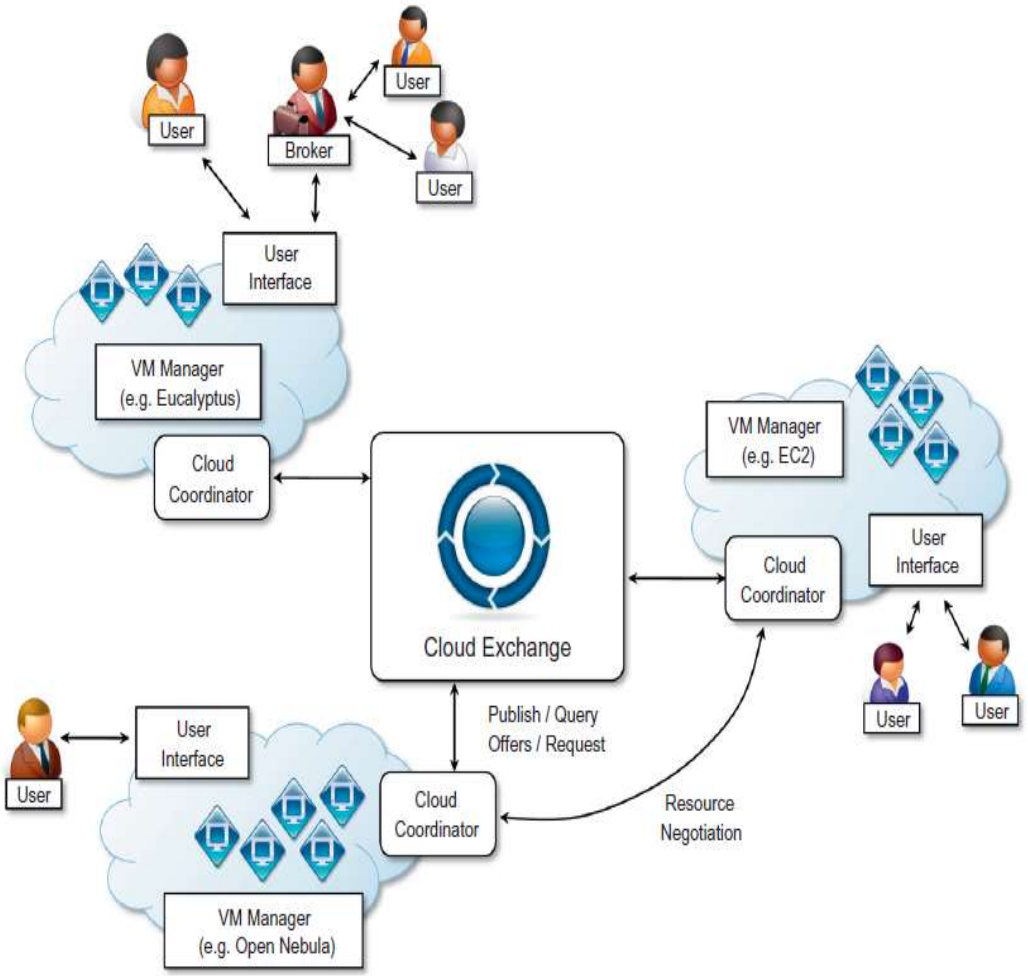


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

CloudCoordinator

Back-end components allows the **Coordinator to learn about the current state of the datacenter** to decide **whether actions from the federation are required or not.**

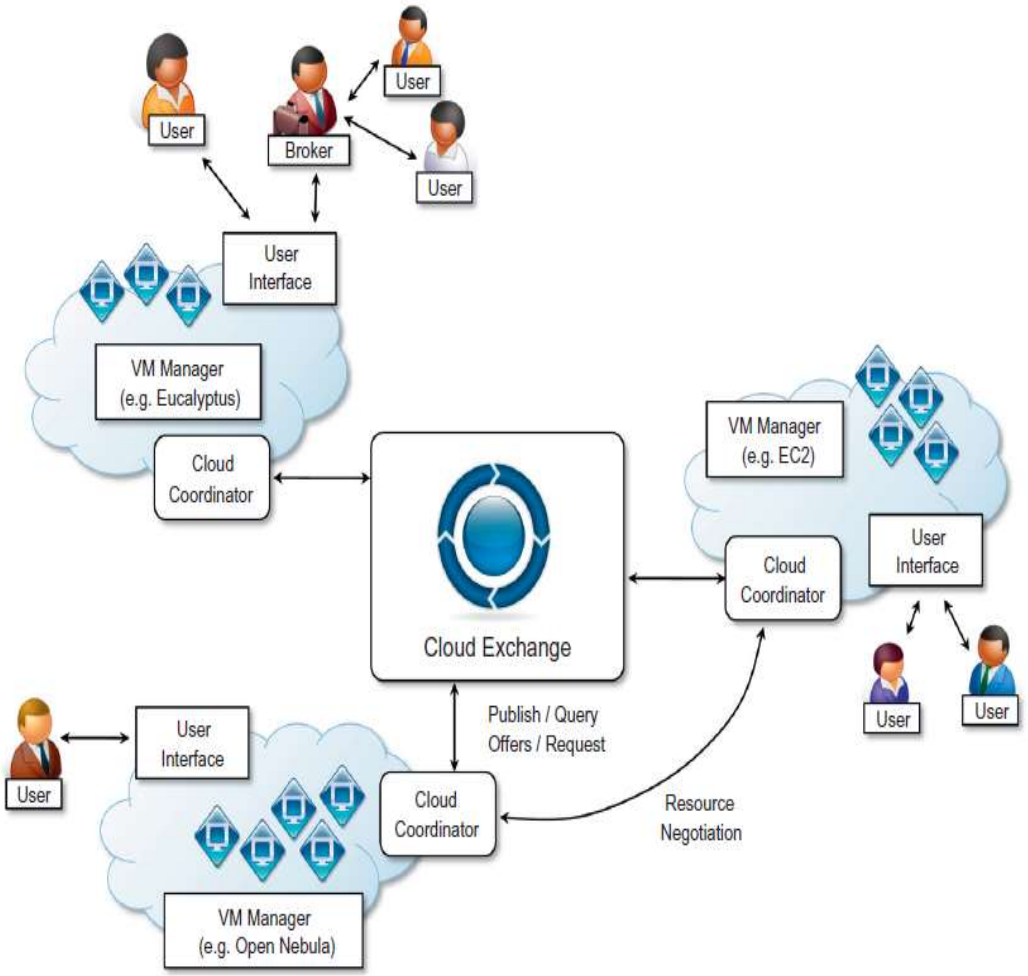


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

CloudExchange

1) Wherever the **Coordinator detects that extra resources are required by the datacenter,**
2) it triggers the **process of discovery of potential providers** (by interacting with the cloud federation).
3) Once **potential providers are discovered and the preferred one is selected, the Coordinator contacts the remote Coordinator and negotiates**

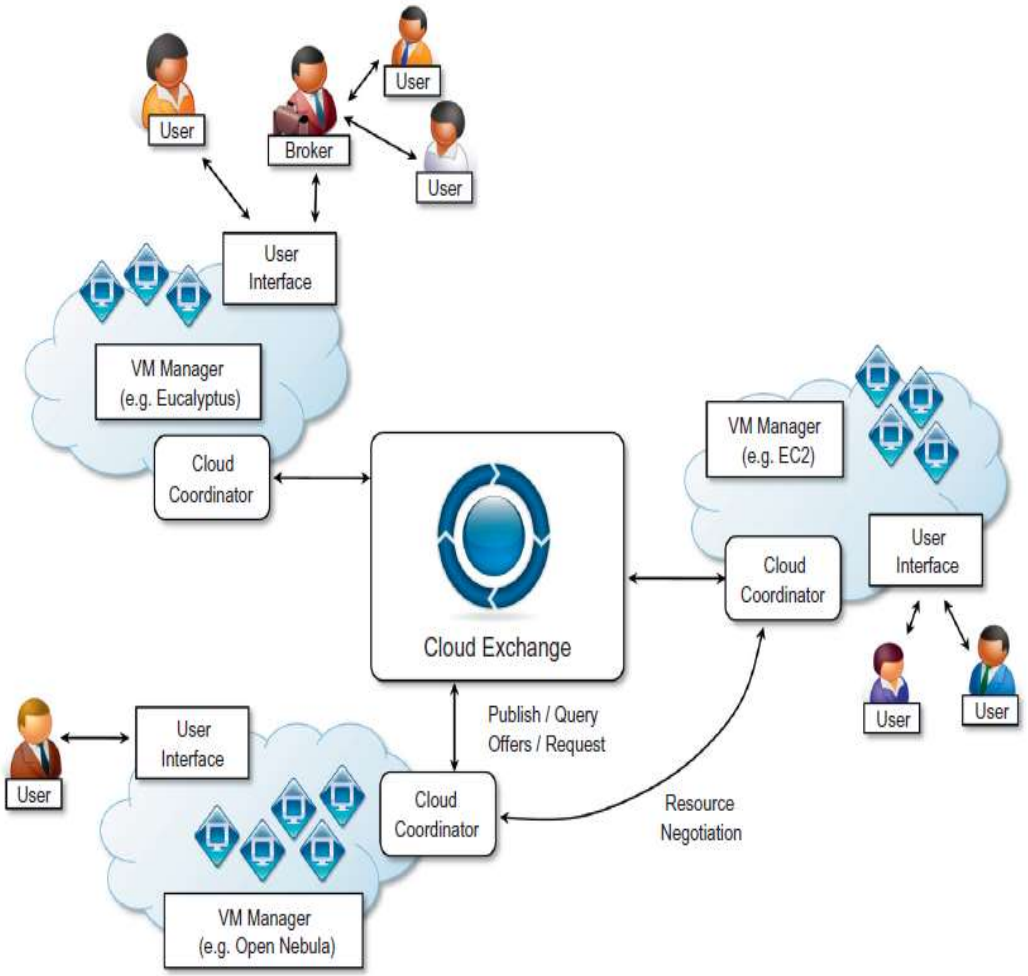


FIGURE 11.12

InterCloud architecture.

InterCloud Architecture

CloudExchange

1) Similarly, when the **Coordinator detects that local resources are underutilized,**
2) They can publish an **offer for resources in the CloudExchange or they can look for matches among requirements registered in the Exchange service.**

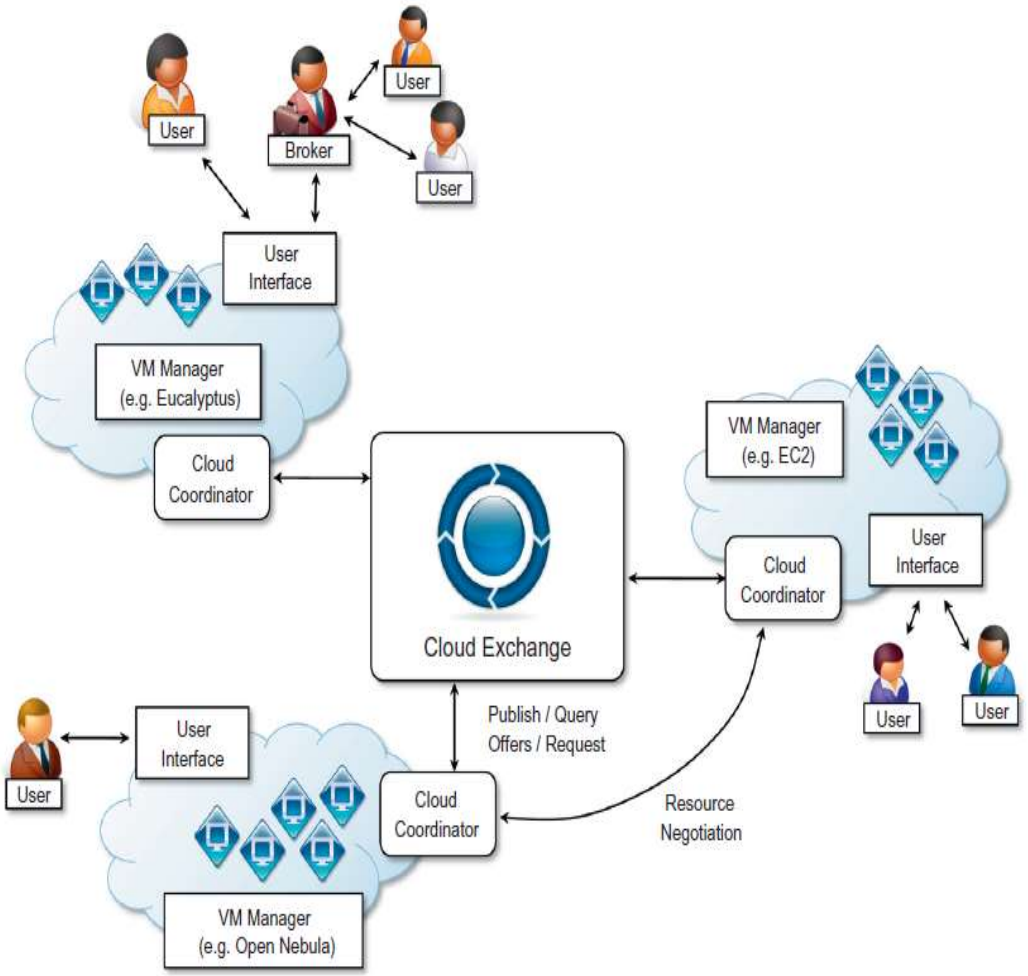


FIGURE 11.12

InterCloud architecture.